



Deep Learning Approaches for Brain Tumor Detection and Classification Using MRI Images (2020 to 2024): A Systematic Review

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Abstract

Brain tumor is a type of disease caused by uncontrolled cell proliferation in the brain leading to serious health issues such as memory loss and motor impairment. Therefore, early diagnosis of brain tumors plays a crucial role to extend the survival of patients. However, given the busy nature of the work of radiologists and aiming to reduce the likelihood of false diagnoses, advancing technologies including computer-aided diagnosis and artificial intelligence have shown an important role in assisting radiologists. In recent years, a number of deep learning-based methods have been applied for brain tumor detection and classification using MRI images and achieved promising results. The main objective of this paper is to present a detailed review of the previous researches in this field. In addition, This work summarizes the existing limitations and significant highlights. The study systematically reviews 60 articles researches published between 2020 and January 2024, extensively covering methods such as transfer learning, autoencoders, transformers, and attention mechanisms. The key findings formulated in this paper provide an analytic comparison and future directions. The review aims to provide a comprehensive understanding of automatic techniques that may be useful for professionals and academic communities working on brain tumor classification and detection.

Keywords Deep learning · Neural network · Brain tumor · Detection and classification

Introduction

Brain tumors, characterized by the abnormal and uncontrolled growth of brain cells, represent the most prevalent form of brain disease [1]. These neoplasms can manifest through both focal and generalized neurological symptoms. Generalized symptoms, indicative of intracranial hypertension, typically include headaches. In severe cases, patients may experience nausea, vomiting, and sixth-nerve palsy. Focal symptoms, on the other hand, can present as hemiparesis or aphasia, depending on the tumor's location. Brain tumors account for 85 to 90% of all primary central nervous system (CNS) tumors [2]. Notably, they are also a significant health concern in pediatric populations, contributing

to considerable mortality rates among children. The complexity of brain tumor classification is underscored by the World Health Organization's (WHO) 2021 Classification of Central Nervous System Tumors, which delineates over 200 types based on location and histopathological features [3]. This intricate categorization poses challenges for detection and classification, particularly for small-sized tumors where differentiation and segmentation become exceptionally difficult. Broadly, brain tumors are categorized as either benign or malignant. Benign tumors, which are non-cancerous, typically grow slowly and have a lower likelihood of recurrence post-treatment. In contrast, malignant tumors are cancerous and possess the ability to metastasize, spreading from the brain to other parts of the body. Given these characteristics, early diagnosis is crucial for timely intervention and effective treatment. While various imaging modalities are employed in brain tumor diagnosis, magnetic resonance imaging (MRI) stands out as the most effective. MRI provides superior contrast of brain anatomical regions compared to CT scans, making it the preferred method for identifying brain tumor malignancy. Traditionally, diagnosis relies

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on radiologists, specialists trained to interpret MRI images and detect anomalies. However, the growing demand for MRI examinations, coupled with a limited number of radiologists, has highlighted the importance of computer-aided diagnosis (CAD) systems. CAD offers several advantages, including reduced workload for healthcare professionals and minimized risk of human error. Moreover, it compensates for the limitations of the human visual system in detecting subtle variations in increasingly complex MRI images. By assisting radiologists in decision-making, CAD plays a vital role in improving time management and enhancing disease detection efficiency. The advent of CAD and artificial intelligence (AI) has opened new avenues for applying innovative algorithms in healthcare, particularly in brain tumor detection. Significant research efforts have focused on incorporating deep learning models for tumor detection and classification. Leveraging high-performance computing devices, these approaches have achieved impressive accuracy levels. Deep learning, a subset of machine learning, has gained prominence in recent years for developing automatic, semi-automatic, or hybrid models capable of detection and classification of brain tumors in less time.

This review includes the most recent papers on various mechanisms of deep learning adopted for the detection and classification of brain tumors from MRI images. What sets this review apart is its pioneering exploration of approaches, including transfer learning, deep autoencoder, transformers, and attention mechanisms and GAN, which have not been previously addressed in the existing literature. Figure 1 shows a taxonomy of the research work in deep learning-based brain tumor diagnosis methods. By systematically evaluating and analyzing a wide range of studies, the primary objectives of conducting this comprehensive review are to:

- Summarize the current state of the art in deep learning-based brain tumor detection and classification.
- Identify emerging trends and advancements in the field.
- Highlight potential research gaps and future research directions.

We have compiled and synthesized findings from an extensive review of about 60 scientific papers. Our investigation encompassed prominent journals within the scientific community. Additionally, datasets used were examined, along with entries from related challenges such as BraTS.

This paper is structured comprehensively as follows: The “**Background**” section provides a background understanding of brain tumors, modalities of imaging, and fundamentals of deep learning. The “**Research Methodology**” section provides the search methodology and outlines the research questions and details the inclusion and exclusion criteria. Within the “**Literature Review**” section, a literature review of the selected studies. The “**Limitations and Gaps**” section provides limitations and gaps, and finally the “**Conclusion**

and Future Direction” section is for conclusion and future directions.

Background

In the following section, we will explore brain tumors comprehensively, covering the definition of brain tumors and their various types. Additionally, we will discuss the imaging techniques used for diagnosing brain tumors, focusing specifically on magnetic resonance imaging (MRI). Moreover, this section will emphasize the basics of deep learning and model architectures.

Brain Tumor

Weighing about 3 pounds in adulthood, the brain is a complex organ that controls behavior and emotions, the seat of intelligence, initiates motor skills, and other cognitive activities. It makes up the central nervous system (CNS) with the spinal cord, which controls voluntary and involuntary movements. It contains blood vessels and nerves, including neurons and glial cells, and is covered by three layers of membrane called meninges. The main parts of the brain are the cerebrum, brainstem, and cerebellum. The cerebrum comprises a convoluted layer of gray matter with neuron somas and white matter which is composed of axons. The brainstem is a very complex structure with a range of different neuron clusters and neural pathways. The cerebellum is shaped like a little brain and contains neurons that control voluntary movement and maintain posture and equilibrium. Deep within the brain, there are other structures such as the pituitary gland, hypothalamus, amygdala, hippocampus, ventricles, and cerebrospinal fluid [4].

A brain tumor is characterized by the abnormal and uncontrolled multiplication of cells within the brain, potentially developing in any part of the brain, skull, or its covering. Although the precise cause of this disease remains unknown, certain risk factors elevate the likelihood of its occurrence, including old age, exposure to radiation, and hereditary conditions. Symptoms of a brain tumor vary based on the affected region of the brain. Common manifestations include nausea, headaches, memory loss, behavior changes, and in severe cases, symptoms may extend to hemiparesis and other neurological deficits [5]. Treatment for brain tumors typically involves a multi-faceted approach, particularly in the early stages. Surgical intervention, chemotherapy, and radiotherapy are common modalities employed to address the tumor directly. Additionally, palliative care may include the use of medications to alleviate symptoms and improve the overall quality of life. In 2021, the WHO updated the nomenclature of brain tumors based on the position, gene, protein, and molecule of tumors [3]. In most cases, a brain tumor receives

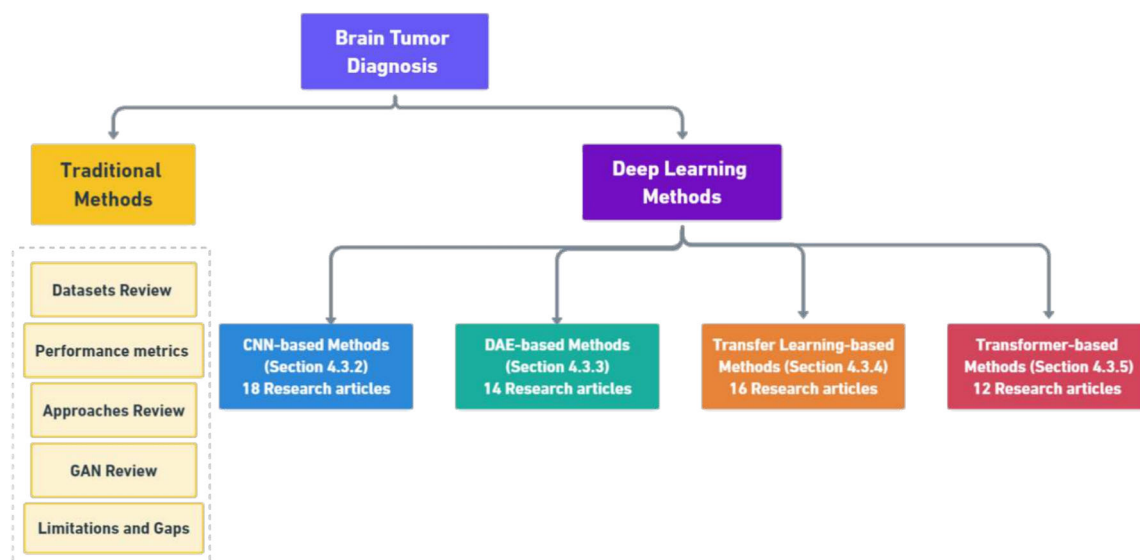


Fig. 1 Our proposed taxonomy of deep learning approaches for brain tumor detection and classification

its name based on the cell type of origin or according to their location (Fig. 2). The most common types are gliomas, meningiomas, and pituitary adenoma (Fig. 3). Hence, most current research on brain tumor classification focuses on these three types.

Gliomas are tumors that arise from glial cells, which consist of 80 to 90% of all brain cells. Glial cells provide supportive functions to neurons, which transmit electrical signals and information throughout the nervous system. However, glioma is a common and the most savage brain tumor, which represents about 42% of all adult brain tumors [6] and approximately 50% in children. Gliomas are usually located in the

deep white matter of the cerebral hemispheres, accounting for 30% of all brain and central nervous system tumors, and 80% of all malignant tumors. The phenotypical makeup of glioma tumors can consist of Astrocytomas, which are tumors developed from astrocyte cells, and Oligodendrogliomas or ependymomas developed from Oligodendrocytes [7]. Astrocytes and Oligodendrocytes are two distinct types of glial cells.

Meningiomas originate from the meninges, which are layers of tissue covering the brain and spinal cord. They stand out as the most prevalent benign intracranial tumors, constituting 10 to 15% of all brain neoplasms, with only a small fraction demonstrating malignant characteristics [8]. Some studies show that meningioma reacts to hormonal changes or trauma [9]. Although it is usually a slow-growing tumor with a good prognosis and it may return after treatment.

A pituitary adenoma is a tumor that forms in the pituitary gland, often referred to as the primary gland of the endocrine system. This gland plays a crucial role in regulating the functions of various endocrine glands throughout the body. Pituitary adenomas are generally characterized by slow growth and are typically classified as benign tumors [9]. They can increase or decrease the production of regulatory hormones. Pituitary microadenomas are smaller than 10 mm in diameter and increase growth hormones. In contrast, macroadenomas are larger than 10 mm in diameter and lead to brain pressure and vision problems. The posterior fossa is a region of the brain that extends from the tentorium to the foramen magnum and is responsible for regulating vital functions. Tumors in this area account for nearly half of all central nervous system tumors in children. The five most common pediatric posterior fossa tumors are diffuse midline glioma, medulloblastoma, ependymoma, atypical

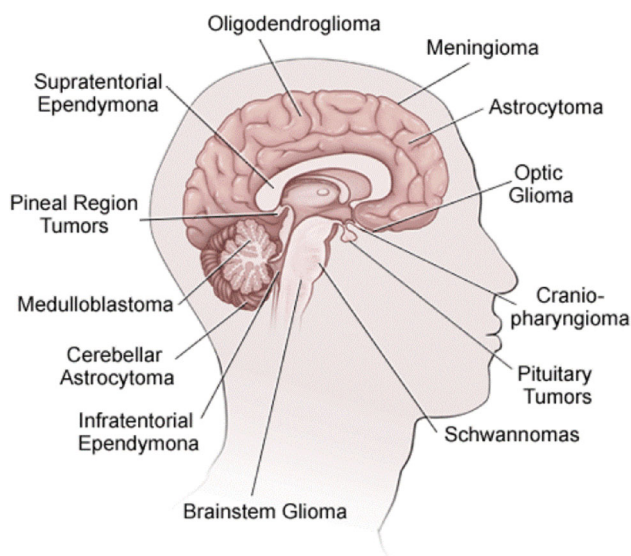


Fig. 2 Brain tumor classification based on tumor location or cell type and origin. The figure shows also the main parts the brain including cerebrum, brainstem, and cerebellum [10]

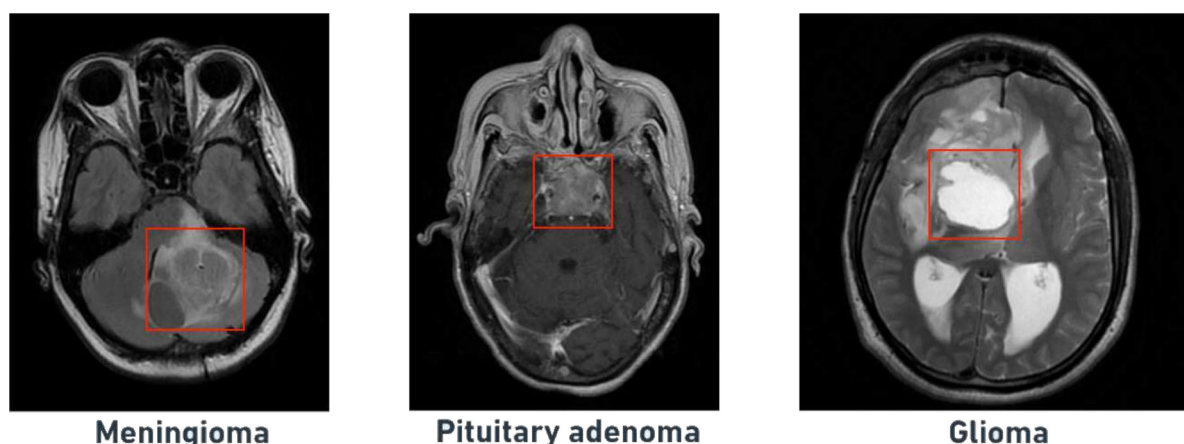


Fig. 3 An example of axial MRI images for common types of brain tumors in which the main difference is the localization of the tumor

teratoid/rhabdoid tumor (AT/RT), and pilocytic astrocytoma [88]. Medulloblastoma is the most prevalent malignant brain tumor in children, predominantly affecting males, with incidence peaks occurring at ages 3 and 9. Ependymoma, the third most common tumor in this region, typically originates from the floor of the fourth ventricle. Pilocytic astrocytoma, the most frequent low-grade tumor in children, is usually diagnosed between the ages of 6 and 8, with no significant gender preference [89].

Furthermore, the WHO 2021 documented brain and spinal cord tumors into 4 grades [3], based largely on how the tumor cells look under a microscope. Lower-grade (grade I or II) tumors tend to grow more slowly with well-defined edges and are less likely to spread into nearby tissues. Hence, they are most likely benign. Higher-grade (grade III or IV) tumors tend to demonstrate swift growth and are more likely to invade surrounding tissues. They can be referred to as malignant and may recur even after intense treatment.

Modalities of Imaging

Early diagnosis of brain tumors plays a crucial role in improving and extending the survival of patients. In parallel, the development of medical imaging technologies has shown significant success in the early diagnosis of not only brain tumors but in all diseases. There are different methods for

capturing medical imaging data including radiography, magnetic resonance imaging (MRI), computed tomography (CT scans), and echography [11]. Among them is magnetic resonance imaging with gadolinium contrast. MRI is a non-invasive medical imaging that uses a strong magnetic field and radio waves to generate high-resolution images without any radiation. It is the first test to diagnose the brain tumor's chemical profile and determine the nature of the lesions. In contrast, Computed tomography can miss structural lesions, particularly in the posterior fossa, or non-enhancing tumors such as low-grade gliomas [12]. Moreover, Positron emission tomography can help detect recurring brain tumors. It provides information on tumor grade and prognosis but they have a relatively poor spatial resolution as compared to an MRI scan [13].

A standard MRI protocol comprises essential sequences, such as FLAIR (Fluid Attenuated inversion recovery), T1, T2, diffusion-weighted imaging (DWI), and T1 with gadolinium contrast [15]. Table 1 describes some relevant properties of these sequences. They are systematically applied in axial, coronal, and sagittal (or volumetric 3D) planes to assess anatomical details, visualize flow voids in vessels, identify associated vasogenic edema, and characterize metabolic peaks (Fig. 4). Although sometimes the only way to make a definitive diagnosis of a brain tumor is through a biopsy. The neurosurgeon performs the biopsy, and the pathologist

Table 1 Properties of different MRI sequences [15]

Modality	Property
T1	Provides high-resolution images of the brain
T2	Provides information about the fluid content of tissue and differentiates between tumor and normal brain tissue
T1-CE	Provides information about blood vessels and the enhancement pattern of the tumor, which aids to diagnose the type of tumor
Flair	Minimizes the bright signal from cerebrospinal fluid and emphasizes small hyper-intense regions of tumors
DWI	Provides information about how easily water molecules are able to diffuse in a region

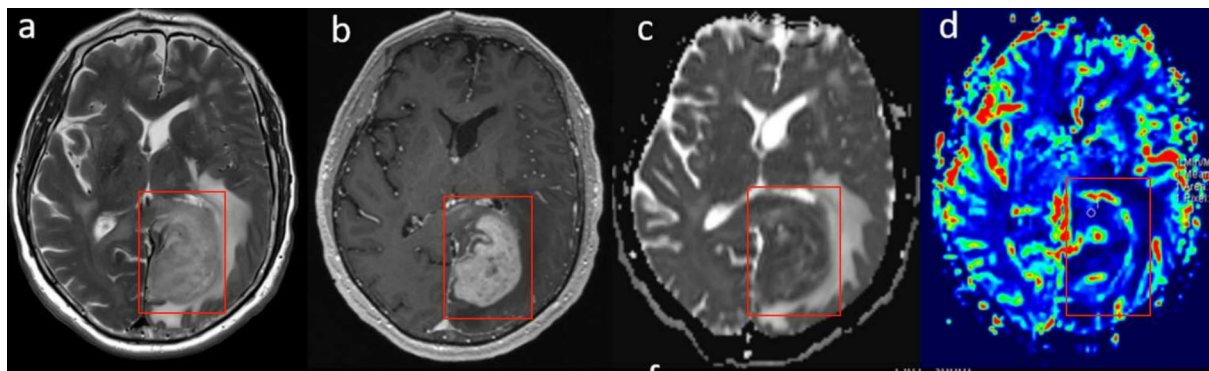


Fig. 4 Four examples of axial MRI images showing a brain tumor in different sequences. **a** T2 in which the tumor appears inhomogeneous and hyperintense. **b** T1 Contrast enhanced shows tumor as heteroge-

neous clearly and hyperintense. **c** DWI tumors appears hypointense. **d** PWI shows low perfusion throughout the lesion [15]

makes the final diagnosis, determining whether the tumor appears benign or malignant and grading it accordingly [3]. Nevertheless, recent research indicates that the integration of multiple MRI modalities enhances the precision of brain tumor segmentation and classification. Hence, it is crucial to account for the significance of the tumor structures across diverse image modalities when devising algorithms for brain tumor detection or segmentation [14].

Deep Learning

Deep learning is a subset of machine learning which is based on applying deep neural networks with three or more layers. These neural networks attempt to simulate the behavior of the biological neural in the human brain and its ability to learn from large amounts of data. Neural networks come in several different forms including the following: convolutional neural networks, deep autoencoders, transformers and attention mechanism, and generative adversarial networks.

In this section, we explain the fundamentals of deep learning and the idea of different deep learning models in the literature, which have been widely applied to medical imaging and particularly for the detection and classification of brain tumors.

Convolutional Neural Network

Convolutional neural networks (CNNs) represent a pivotal innovation in the field of artificial intelligence, particularly in computer vision. They are a type of artificial neural network and inspired by the central nervous system of the human brain. With their abilities to automatically learn features, CNN can handle 2D and 3D MRI images and provide segmentation of brain tumors, detection of abnormalities, and classification of types of tumors.

CNN transforms an input image into a feature map, by assigning trainable weights with biases to different aspects

of the image through a series of convolutional and pooling layers, and then processed through multiple convolutional and pooling layers to produce a predicted output. Convolutional layers play a crucial role in feature extraction from a given image, identifying essential patterns like edges and lines. This process involves utilizing a convolution operation with a moving window matrix known as a filter or kernel to extract these distinctive features. The convolution expression is defined as:

$$C(i, j) = \sum_m \sum_n I(i + m, j + n) \cdot K(m, n) \quad (1)$$

Where:

$C(i, j)$ is the value at position (i, j) in the output feature map. $I(i+m, j+n)$ represents the pixel value at position $(i+m, j+n)$ in the input image.

$K(m, n)$ is the weight (or kernel coefficient) at position (m, n) in the kernel.

The double summation involves iterating over all possible positions (m, n) in the kernel.

The output of each convolutional layer forms a feature map which represents the spatial distribution of features. Because images are far from linear and convolution is a linear operation, an activation function introduces non-linearity to the activation map. There are several types of non-linear activation functions like ReLU, Tanh, sigmoid, softmax, Leaky ReLU, etc.

To reduce the spatial dimensions of the feature maps, pooling layers are commonly used. These layers contribute to the networks to retain the most prominent features within that region and extract important information. For instance, these layers help to reduce computational complexity and decrease the number of parameters leading to a fast and efficient training.

Moreover, fully connected layers then establish connections between all neurons in the previous layers and all the neurons in the next layer. It can be computed using matrix

multiplication followed by a bias effect. These layers serve as the combination of the network's ability to discern intricate patterns and relationships within the data, contributing significantly to the classification process and decision-making. Figure 5 illustrates a standard architecture of CNN. In addition, integrating batch normalization into CNN architectures enhances independent learning within each network layer. This technique scales input activation layers during the normalization process, facilitating effective learning and acting as a regularization mechanism to prevent overfitting. Furthermore, Dropout layers address overfitting concerns by selectively removing a percentage of neurons during training, fostering a more compact model that generalizes well to new data. When a dropout rate of 0.3 is applied, 30% of the nodes in the neural network are randomly dropped out, contributing to a more robust and adaptable model.

CNNs offer several advantages that make them particularly powerful for medical image analysis. In fact, CNNs automatically learn relevant features directly from the data through their convolutional layers, unlike traditional methods. This eliminates the need for extensive domain knowledge and reduces bias in feature selection.

In addition, with alternating convolutional and pooling layers, CNN captures spatial relationships and local patterns within images. This makes them exceptionally well suited to tasks like brain tumor detection and classification often achieving state-of-the-art performance.

Deep Auto-encoder (DAE)

The concept of autoencoders has evolved over time however, one influential work often cited in the context of autoencoders is Geoffrey Hinton and Ruslan Salakhutdinov's paper titled "Reducing the dimensionality of data with neural networks," which was presented at the Conference on Neural Information Processing Systems (NeurIPS) in 2006 [16]. In this paper, Hinton and Salakhutdinov proposed a method for learning a non-linear dimensionality reduction, which can be considered a precursor to the autoencoder architecture. They introduced the idea of training a neural network to encode and decode data, showcasing its ability to learn efficient representations of input data in a lower-dimensional space.

An autoencoder comprises three essential components. The encoder is responsible for transforming the input data into a significantly smaller encoded form, while the bottleneck stores this compressed knowledge. The decoder then reconstructs the original input from its encoded form. This tripartite structure ensures that the autoencoder effectively captures and retains essential features during both the compression and reconstruction phases. Figure 6 illustrates a general architecture of a DAE.

In the context of medical imaging, autoencoders prove invaluable for tasks such as denoising, dimensionality reduc-

tion, and anomaly detection. Moreover, autoencoders offer several advantages, notably in the context of brain tumor imaging. They excel in denoising MRI images, improving image quality for precise analysis. Additionally, the ability to learn low-dimensional representations of complex neuroimaging data enhances visualization and interpretation, crucial for identifying unique features related to specific neurological conditions.

Transfer Learning

Transfer learning is a class of deep learning that has demonstrated its potential for detecting brain tumors in recent years. Transfer learning allows the use of a previously trained CNN architecture. The popular pre-trained networks are AlexNet, VGGNet, GoogLeNet, ResNet, and InceptionNet. These models are used for brain tumor segmentation, detection, and classification. Therefore, using the extracted features of the pre-trained networks as a backbone to train the first stage of the network is much easier and faster than training it from scratch [17]. Figure 7 shows the architectures of the following pretrained models.

- **AlexNet:** Introduced in 2012 and crafted by Alex Krizhevsky et al. [18], AlexNet revolutionized convolutional neural networks by being the first to leverage GPU for enhanced performance. AlexNet has five convolutional layers, three max-pooling layers, two normalization layers, two fully connected layers, and one softmax layer. Thus, a nonlinear activation function ReLU and Convolutional filters are used in each convolutional layer. AlexNet's architecture incorporated 60 million parameters, marking a milestone in the efficient use of GPUs and influencing subsequent developments in deep learning.
- **VGG16:** or Visual Geometry Group 16 was presented by the Visual Geometry Group at the University of Oxford in 2014. VGG16 architecture includes 16 convolutional layers, 5 max-pooling layers, 2 fully connected layers, and one softmax layer. Hence, for classification, a flattened layer, 2 batch normalization layers, 3 dense layers, and 2 layers of dropout which are substituted for fully connected layers [19].
- **ResNet-50:** a cutting-edge deep convolutional neural network architecture that was developed by Microsoft Research in 2015. Its main concept is to introduce an "identity shortcut link" that bypasses one or more layers. One of its key innovations is the use of residual connections, which allow the network to learn a set of residual functions that map the input to the desired output. It comprises 50 layers that enable the network to learn much deeper architectures without suffering from the problem of vanishing gradients [20]. The authors in [21] claimed that this is one of the deepest architectures.

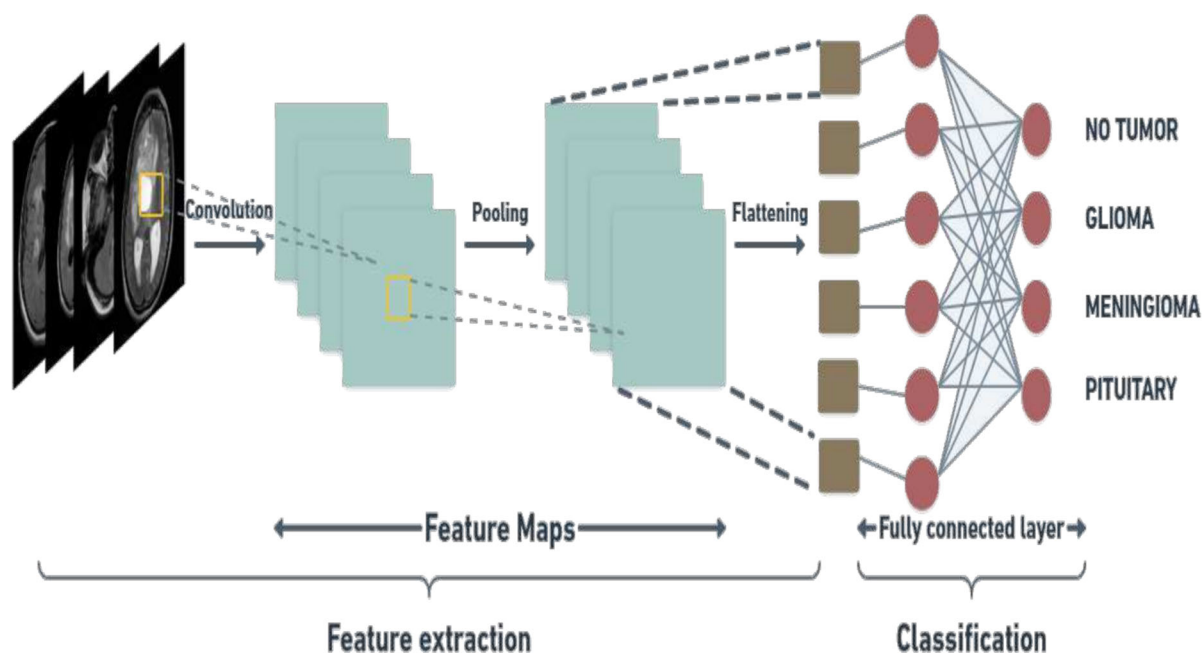


Fig. 5 Standard architecture of CNN. The diagram shows the stages involved in CNN, including convolution, pooling, feature maps generation, flattening, and final output of classification. Best viewed in color

- **DenseNet:** deep neural network design known as Dense Convolutional Network was first presented in 2017 by Huang et al [22]. DenseNet is an effective solution to the problem of vanishing gradients and has a significantly reduced number of parameters. In a DenseNet architecture, each layer is connected to every other layer. For each layer, the feature maps of all the preceding layers are used as inputs, and its own feature maps are used as input for each subsequent layer [23].
- **InceptionV3:** a superior version of the basic model Inception V1 which was introduced as GoogLeNet in 2014 by a team at Google. It is one of the more popular deep learning models based on CNN designs. In total, the inception V3 model is made up of 42 layers which is a bit higher than the previous inception V1 and V2 models [24].

While transfer learning offers a promising solution to the challenges faced by traditional CNNs, it is essential to

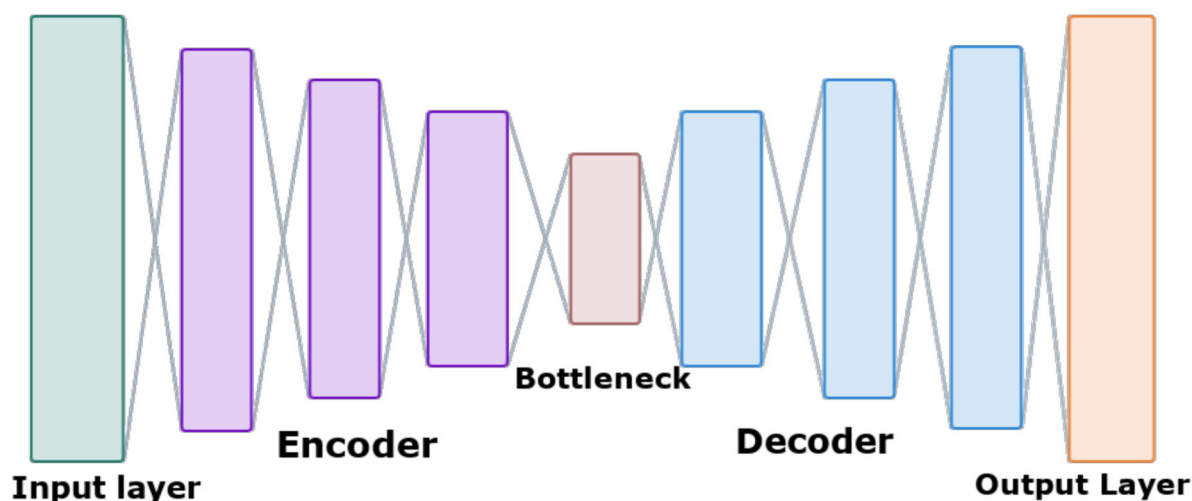


Fig. 6 General architecture of a DAE. The encoder efficiently captures features through convolutional and pooling layers, leading to a bottleneck layer for compression. The compressed representation is then decoded through mirrored layers in the decoder, resulting in the final output layer. Best viewed in colors

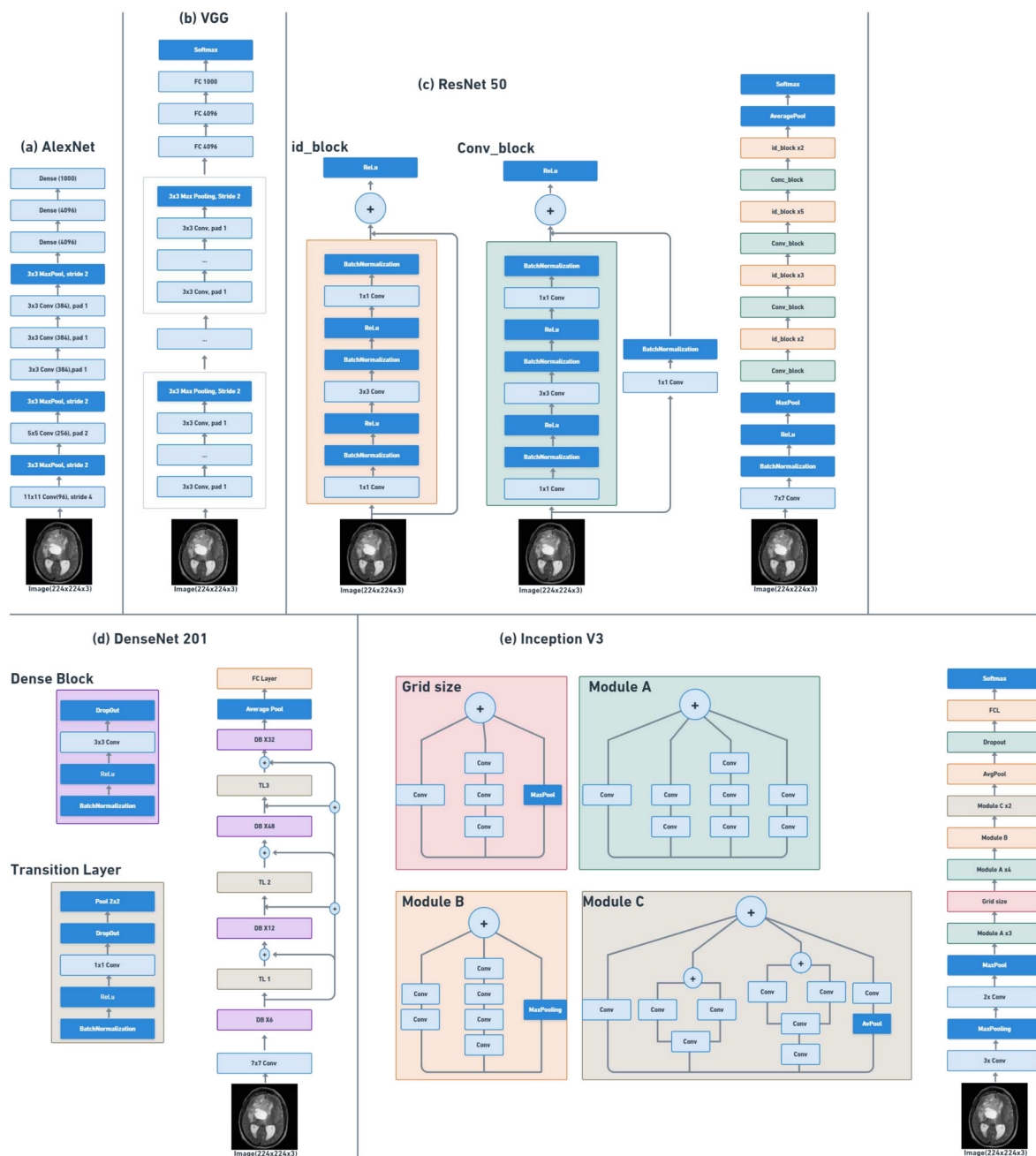


Fig. 7 Architectures of pre-trained models: **a** AlexNet has five convolutional layers, three max-pooling layers, and three fully connected layers. **b** VGG Network has an initial segment with convolutional and pooling layers, followed by a segment with fully connected layers. The convolutional segment integrates multiple VGG block modules in a successive manner. **c** ResNet50 follows VGG's 3×3 convolutional layer structure and includes a residual block with two 3×3 convolu-

tional layers. Batch normalization and ReLU activation functions are applied after each convolutional layer. **d** DenseNet 201 is named for its dense connectivity in the dependency graph among variables. It includes dense blocks and transition layers. **e** Inception V3 is a CNN architecture with enhancements such as label smoothing, factorized 7×7 convolutions, and an auxiliary classifier. The Inception Module is at the heart of Inception V3. Best viewed in color

acknowledge its drawbacks, particularly in the context of brain tumor detection. One notable limitation is the potential domain mismatch between the pre-trained model's source task and the target task of identifying brain tumors. The

intricate and nuanced nature of medical imaging datasets, combined with variations in imaging modalities and patient demographics, can lead to a misalignment that hampers transfer learning effectiveness. Additionally, transfer learning

may struggle with capturing specific intricacies inherent to brain tumor detection, demanding fine-tuned adjustments for optimal performance. As we navigate through these limitations, the transition to the transformative role of transformers in addressing these challenges becomes evident. Transformers, with their attention mechanisms and ability to capture complex relationships in data, represent a compelling shift in paradigm for enhancing the accuracy and efficiency of brain tumor detection models.

Transformers and Attention Mechanism

The transformer architecture is introduced for the first time in the paper “Attention is All You Need” by Vaswani et al. in 2017 for natural language processing tasks [25]. It has achieved remarkable success in machine translation, text classification, and language modelling. Recently, transformer-based architectures have rapidly appeared in computer vision and have shown promising performance in medical image analysis. Its self-attention mechanism allows for capturing contextual information between positions in the input sequence. These attention-based transformer models have become an attractive solution due to their ability to encode long-range dependencies and learn highly effective feature representations. Moreover, inspired by this, Dosovitskiy et al. (2020) introduced the classical transformer into computer vision called Vision Transformer (ViT) [26]. Specifically, ViT serves as a deep-learning framework designed specifically for image classification tasks. It utilizes tokenized image patches and employs the self-attention mechanism to effectively capture global context. The architecture initiates the process by partitioning an image into a sequence of patches. Subsequently, these patches are then transformed into a 1D vector, which is then fed into a series of Transformer blocks. Each Transformer block comprises multiple attention mechanisms, and their weights are combined across various layers. Ultimately, the output layer is utilized for the classification procedure. In a more detailed explanation, prior to entering the transformer block within the ViT architecture, the data undergoes several layers, including Patching, Positional Encoding, and the generation of Classification tokens [26].

Research Methodology

In the following section, we delineate our research methodology, providing a structured approach to address our key objectives. We begin by formulating targeted research questions that guide our exploration. To navigate the expansive landscape of existing literature, we deploy carefully curated search keywords. Therefore, we establish clear inclusion and

exclusion criteria of the investigated studies. Following the PRISMA framework, our data extraction process is systematic and transparent, systematically synthesizing key findings and methodological nuances.

Figure 8 shows the process of selecting studies for the research which is divided into three steps:

- Identification: searching for articles using specific search terms across various databases.
- Screening: applying exclusion criteria and reviewing titles and abstracts to exclude those judged irrelevant.
- Inclusion: conducting a comprehensive study of the selected articles to prepare the final set of main articles.

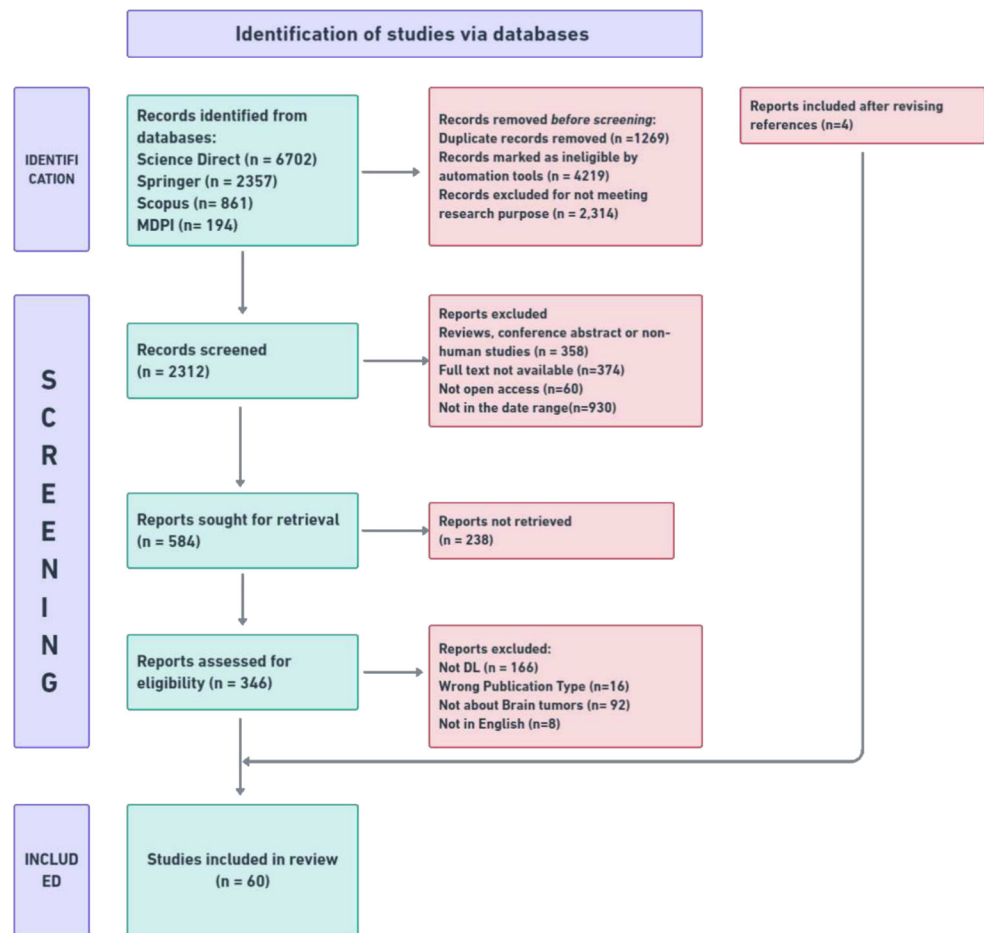
Research Questions

In this study, a rigorous and systematic approach is adopted, grounded in the principles of a systematic literature review (SLR). This methodological framework is chosen to ensure a comprehensive and transparent exploration of the existing studies on deep learning applications for brain tumor detection and classification. The research questions guide the entire investigation, directing the focus toward key aspects of deep learning applications in brain tumor diagnosis. Therefore, the primary focus of this study is to address the research questions as follows:

- RQ1: What are the commonly used datasets for diagnosing brain tumors based on deep learning?
- RQ2: What types of deep learning algorithms are commonly employed for diagnosing brain tumors?
- RQ3: What is the accuracy achieved by each of these algorithms?
- RQ4: What resources and software tools are commonly used in this context?
- RQ5: What challenges and highlights are associated with these methods?

The SLR methodology involves a series of systematic steps. First, a comprehensive search is conducted to identify studies relevant to the research questions. Keywords and search terms are carefully chosen to cast a wide net over the existing literature. Then, rigorous inclusion and exclusion criteria are then established to ensure that the selection of studies aligns with the research objectives. This step is crucial for maintaining the quality and relevance of the included literature. While data Extraction includes details on datasets, deep learning algorithms, accuracy metrics, resources, and challenges. Finally, the extracted data is synthesized and analyzed to draw meaningful conclusions.

Fig. 8 PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) flowchart of selecting studies



Search Keywords

The search terms were merged into a search string utilizing the Boolean operator “OR” to incorporate synonyms and alternative spellings, and the Boolean operator “AND” to connect the keywords. This combination of terms formed the basis for the following search query:

Query 1: (“Deep learning” OR DL) AND “brain tumor” AND (classification OR detection)

Query 2: (“Convolutional neural network” OR CNN OR “neural network”) AND “brain tumor” AND (detection OR classification)

Moreover, this article has been systematically explored across various scientific databases, and references from these sources have been formally acknowledged. The use of databases plays a crucial role in scholarly articles, and for this study, specific electronic databases have been meticulously chosen. As illustrated in Fig. 9, the electronic databases employed in this research include:

- ScienceDirect - Elsevier (<http://www.elsevier.com>)
- Springer Link (<http://www.springerlink.com>)
- MDPI (<https://www.mdpi.com/>)

- Scopus (<https://www.scopus.com/>)

Inclusion and Exclusion Criteria

To assess the article’s purpose, we conducted a meticulous selection process from a comprehensive pool of articles identified using search queries. This selection procedure adhered to predefined criteria, as outlined in Table 2, including both exclusion criteria (EC) and inclusion criteria (IC).

In the process of study selection, our aim was to ensure the inclusion of high-quality and valuable research that aligns closely with the objectives of this systematic review. To achieve this, we established explicit inclusion criteria that prioritize articles published in reputable journals known for their rigorous peer-review processes and contributions to the field. Specifically, we focused on articles published in journals with high-impact factors and distinguished journal rankings (e.g., Q1). This approach was taken to enhance the reliability and validity of the evidence synthesized in this review.

While acknowledging that impact research can be found in a variety of journals, our intent was to maintain a stringent standard for the inclusion of studies to ensure the robustness and relevance of the synthesized evidence. The criteria set

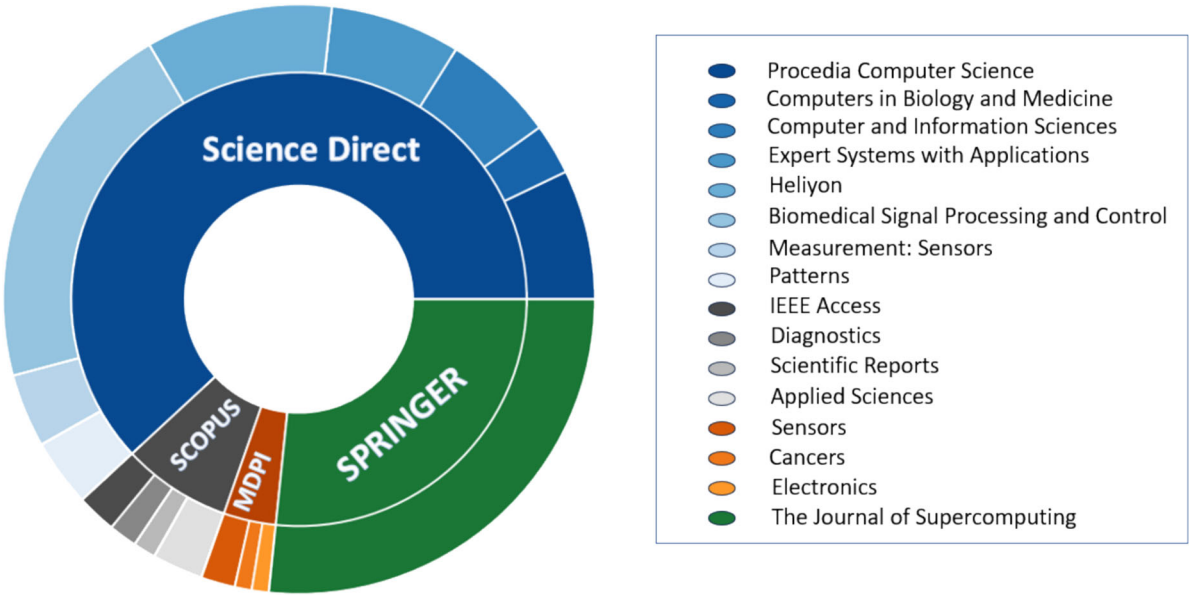


Fig. 9 Distribution of the included papers divided by publishers and journals. The distribution highlights the prevalence of papers from publishers, with a significant presence from ScienceDirect, followed by

contributions from Springer, Scopus, and MDPI. Overall, this distribution highlights the diverse sources that contribute to the comprehensive scope of the literature review. Best viewed in colors

forth aimed to prioritize studies with a strong methodological foundation and meaningful contributions to scientific discourse. It is important to note that these criteria were applied transparently and consistently across the selection process.

The exclusion of studies from journals with lower impact factors or rankings does not diminish their potential value; rather, it reflects a deliberate decision to focus on a subset of articles that meet our stringent quality standards. Our commitment to transparency extends to providing a clear rationale for these inclusion criteria, and we acknowledge that other valuable research may exist beyond the predefined thresholds.

This approach contributes to the overall rigor of our systematic review and underscores our dedication to extracting data from studies that not only meet high-quality standards but also align closely with the objectives and scope of this review. Algorithm 1 provides a structured framework for sys-

tematically acquiring and filtering research papers based on predefined criteria.

Data Extraction

In the course of this research endeavor, the data extraction and article filtering processes were facilitated through the use of Rayyan (www.rayyan.ai/) a systematic review tool designed to streamline the literature review process. Rayyan allows for the efficient organization and management of research articles by providing advanced features such as screening, tagging, and collaboration tools. Utilizing Rayyan’s filters, we refined our dataset by systematically excluding or including articles based on predetermined criteria and ensuring a focused and relevant set of studies for analysis. This tool played a pivotal role in expediting the screening process and enhancing the precision of our research.

Table 2 Inclusion and exclusion criteria for article selection

Category	Criteria
Inclusion	Studies related to brain tumor detection or classification Articles published between 2020 and 2024 Research articles published in journals with a high impact factor and citation Articles in English and open access
Exclusion	Studies not related to human brain tumors Research using CT scans and histopathological images Articles on other cancers or anatomical regions Studies using methods other than deep learning MRI images post surgery with the goals of treatment response assessment

Additionally, to manage and organize the extensive references encountered during the literature review, we employed Zotero as our reference management tool. Zotero's capabilities for citation management and collaboration complemented Rayyan's functionality, contributing to a comprehensive and well-organized research workflow. The combination of Rayyan and Zotero proved instrumental in navigating the complexities of literature review and data extraction for this study. To enhance the organization of our review process, we employed a table (Fig. 16 presented in Appendix 1) that delineates the steps to facilitate a deeper understanding and comprehensive review of the included papers.

Algorithm 1 Data extraction algorithm from research databases.

```

1: Research databases  $\leftarrow$  DB
2: Published year range  $\leftarrow i$ 
3:  $N \leftarrow 5$   $\triangleright$  Number of research databases
4:  $I \leftarrow 1$   $\triangleright$  Starting value
5: for  $I \leq N$  do
6:   Search keywords  $\leftarrow$  deep learning, brain tumor, classification,
     detection
7:   if  $i$  in (2020–2024) AND DB in [ScienceDirect, MDPI, Springer,
     Scopus] then
8:     Search for papers using the keywords
9:   end if
10:  if  $Papers \geq 0$  then
11:    Inclusion criteria  $\leftarrow$  IC1, IC2, ..., ICn
12:    Exclusion criteria  $\leftarrow$  EC1, EC2, ..., ECn
13:  end if
14: end for

```

Literature Review

This section navigates through the landscape of publicly available datasets and metrics used in the field of brain tumor detection and classification. Additionally, this section delves into both traditional and deep learning methodologies. A meticulous exploration of the datasets utilized in these paradigms provides critical insights into the performance and limitations of detection models.

Review of Available Datasets

Researchers often use specific datasets that are considered benchmarks in their respective domains. These datasets are well established, widely recognized, and serve as a standardized set of examples for evaluating algorithms or models. Table 3 determines the publicly available and commonly used datasets for brain tumor detection. Two noteworthy datasets that play pivotal roles in this research are the Multimodal Brain Tumor Segmentation Challenge (BraTS) and Figshare.

The Multimodal Brain Tumor Segmentation Challenge (BRATS) stands out as a significant contributor to the field, providing datasets considered among the most challenging

MRI datasets issued over various years. Specifically, the latest version of the BRATS training dataset, released in 2023, comprises 228 pediatric high-grade glioma cases. Utilizing NIfTI files (.nii.gz) for segmentation and DICOM (.dcm) files for classification, BRATS offers a comprehensive set of data reflecting the complexity of real-world brain tumor cases.

Figshare, a prominent digital repository platform, serves as a valuable resource for hosting diverse research materials, including datasets pertinent to brain tumor research. The Figshare dataset included in this study is characterized by .mat files. Figshare's open-sharing approach and secure accessibility make it an essential platform for researchers seeking a collaborative and transparent space for scholarly outputs.

These datasets, alongside others such as BrainWeb, Harvard Whole Brain Atlas, Kaggle, IXI Brain Development Dataset, IBSR, RIDER, and TCIA, collectively form a rich and varied pool of resources. Each dataset contributes unique attributes, from synthetic brain images with known ground truth (BrainWeb) to real brain images sourced from different hospitals (IXI Brain Development Dataset). The inclusion of both challenging datasets like BRATS and versatile repositories like Figshare underscore the comprehensive approach taken to gather a diverse set of data for the research at hand. Figure 10 illustrates the distribution of articles for each datasets. It is evident that Figshare is the most frequently used followed by Kaggle and BraTS.

Performance Metrics

In line with the constructive feedback principle inherent in constructing DL models, these models receive feedback based on metrics. Evaluation metrics play a crucial role in elucidating the performance of the model. A comprehensive list of commonly used metrics is provided in Table 4. Among these metrics, accuracy stands out as the most frequently utilized one. Given the primary objective of brain tumor classification, accuracy serves as a key indicator, providing insights into the proportion of correct predictions made by the model.

Review of Deep Learning Approaches

In the following section, we will discuss the use of deep learning approaches for the detection and classification of brain tumors. We will start by examining traditional methods including machine learning. Subsequently, we will delve into specific deep learning approaches used for detection and classification of brain tumors.

Traditional Methods

Before delving into deep learning approaches, a comprehensive understanding of traditional methods. Familiarity

Table 3 Summary of the publicly available datasets

Datasets	Source	Description	File format
MICCAI BRATS	[77]	Stands for the Multimodal Brain Tumor Segmentation Challenge. The BRATS datasets are considered to be the most difficult MRI datasets issued at different times throughout the years. The latest version (2023) of the BRATS training dataset has 228 pediatric high-grade glioma [76]	NIFTI files (.nii.gz) for Segmentation, DICOM (.dcm) files for classification
BrainWeb	[78]	A synthetic brain MRI dataset developed to facilitate research and evaluation in the domain of medical imaging. It provides realistic simulated brain images with known ground truth, making it a valuable resource for tasks such as image segmentation, registration, and other image processing applications	MINC and raw(binary data)
Figshare	[79]	A digital repository platform designed for the open sharing of scholarly outputs. Figshare provides a secure and accessible space for hosting diverse research materials, including datasets, figures, and supplementary files	.mat files
Harvard Whole Brain Atlas	[80]	A website that provides dozens of real brain images, both normal and diseased	mpg files
Kaggle	[81]	A data science competition platform and online community of data scientists and machine learning practitioners under Google LLC	Jpg files
IXI Brain Development Dataset	[82]	The data has been collected as part of the project Information eXtraction from Images at three different hospitals in London. It contain 600 MRI images T1, T2, PD weighted and diffusion [83]	NIFTI files
IBSR	[84]	The internet brain segmentation repository: a world wide web resource providing access to magnetic resonance brain image data and segmentation results	NIFTI files
RIDER	[85]	Publicly available on The Cancer Imaging Archive (TCIA) contains MRI imaging T1-CE 3D of 19 patients with recurrent glioblastoma	DICOM files
TCIA	[86]	Publicly accessible repository that hosts a vast collection of medical images and related data, with a primary focus on cancer imaging from 20 subjects with primary newly diagnosed glioblastoma	DICOM files

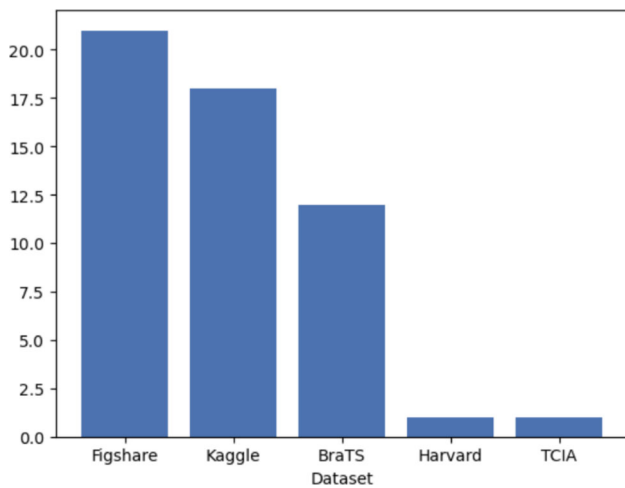


Fig. 10 Distribution of articles for each dataset. Figshare and Kaggle emerge as the most frequently utilized datasets, with BraTS following closely behind

with these conventional methodologies forms a foundational knowledge base for advancing and appreciating the innovations brought by deep learning in the context of brain tumor diagnosis. The most commonly used steps to diagnose brain tumors using deep learning are pre-processing, segmentation, feature extraction, and classification and are presented in Fig. 11 with some examples of techniques used. Preprocessing is a series of transformations or mathematical morphology applied to an initial image for improving the image quality. The efficiency of MRI images is influenced by artifact acquisition such as intensity non-uniformity and Gaussian noise. Traditional preprocessing techniques, such as median, Gaussian, and Gabor filters, play a vital role in refining raw data before input into deep learning models, enhancing performance in image classification and segmentation tasks. In the context of MRI image quality improvement, de-noising, intensity correction, skull stripping, and normalization are crucial preprocessing steps. Raja

[27] used in the pre-processing stage the non-local mean filter for denoising purposes.

Feature extraction involves retrieving statistical measurements from brain MR scans, including shape, structure, and wavelet information. The Gray-Level Co-occurrence Matrix (GLCM) is commonly utilized to extract second-order statistics, providing essential features for accurate brain tumor classification. It is defined as a two-dimensional histogram of gray levels for a pair of pixels, which are separated by a fixed spatial relationship. Studies employ GLCM-Tamura [28], discrete wavelet transform (DWT), and Gabor filtering [29]. Moreover, automatic features produced by deep learning techniques like convolutional neural networks, ResNet, and Capsule Networks have shown success [30].

Moreover, segmentation in medical images is a method of dividing pixels. A tag is given to each pixel in the image, and pixels with the same label have similar properties which consist the target area usually a lesioned region. This task is in order to detect and separate the target from the background and healthy tissues. Segmentation of a brain tumor is a challenging task and needs a skilled person having knowledge of tissues. Thus, it is an erroneous task because of low-level visual information that causes mis-segmentation and tissue loss even when performed by a skilled individual [31]. Consequently, computer software performs segmentation and eliminates the need for manual human interaction [32] using various automated techniques such as thresholding, region-based segmentation, region growing, watershed, etc. Thresholding involves using one or more thresholds to classify regions of the image based on their intensities [33]. By setting a specific intensity threshold, pixels with values above or below are classified as part of the tumor or normal tissues. This principle forms the foundation for various image enhancement techniques that help to extract important features from an image for further analysis. Various thresholding methods exist, such as global thresholding or adaptive thresholding. Shemanto et al. [34] used a technique of raising

Table 4 Summary of the commonly used metrics

Metrics	Meaning	Mathematical representation
Accuracy	The proportion of correct prediction to the total number of predictions	$\frac{TP+TN}{TP+TN+FP+FN}$
Specificity	The proportion of negative cases correctly identified as negative	$\frac{TN}{TN+FP}$
Precision	The proportion of true positive cases that are correctly identified from the total positive cases	$\frac{TP}{TP+FP}$
Recall	The fraction of positive cases correctly identified as positive. Also known as sensitivity	$\frac{TP}{TP+FN}$
Dice similarity coefficient	The harmonic mean between precision and recall value. Also known as F Measure and F Score	$\frac{2TP}{2TP+FP+FN}$
Geometric mean	Measures the trade-off between the sensitivity and specificity	$\sqrt{\text{Sensitivity} \times \text{Specificity}}$
Intersection-overunion (IoU)	The ratio of the intersection of the reference and classified samples with the union of the two groups. Also known as JACCARD Similarity Index (JSI)	$\frac{TP}{TP+FP+FN}$

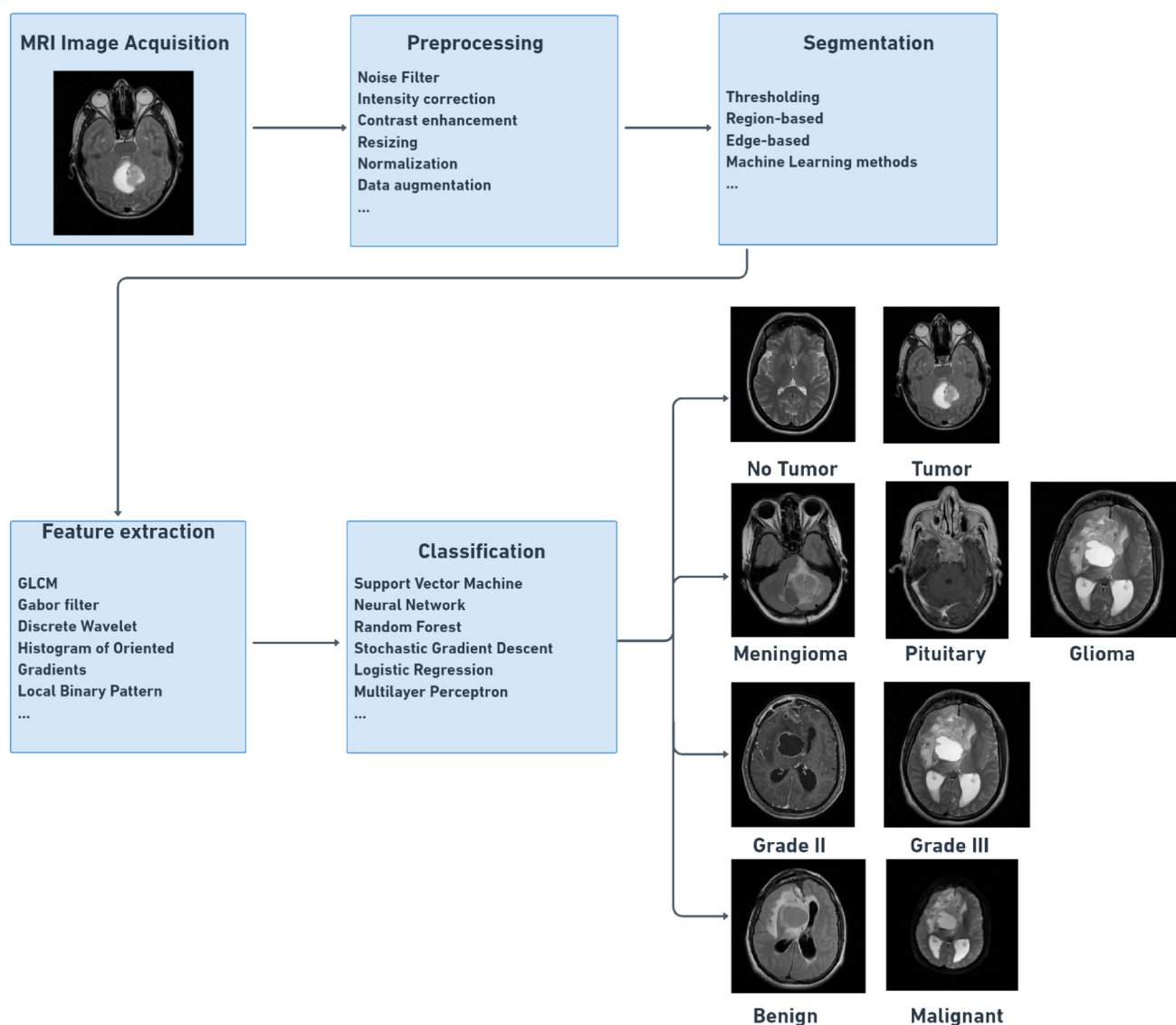


Fig. 11 Commonly used pipeline for brain tumor classification. The workflow includes essential steps such as preprocessing, segmentation, feature extraction, and classification

the threshold after acquiring the initial value calculated by the gray-thresh function. Hence, various increases were required at various threshold levels. This thresholding approach was used to eliminate brain malignancies from 2D MRI images. However, this method has some drawbacks such as low convergence rates due to being confined in local minimums [35]. Other techniques are utilized based on region such as region growing and watershed. Their main idea comes from that pixels with similar intensities are grouped together. Thus, these techniques usually suffer from over segmentation and under segmentation.

To overcome these limitations, numerous machine learning methods are available, including soft computing techniques, atlas-based, clustering namely K-means clustering and fuzzy C-means clustering. Raja [27] used the Bayesian

fuzzy clustering in the segmentation process. The BFC model is composed of data likelihood distribution called the Fuzzy Data Likelihood (FDL). It categorizes the edema and core tumor areas and produces the fragments for mining features. Thus, machine learning has demonstrated some limitations such as overfitting. Consequently, many researchers developed models using deep learning techniques to overcome this issue. In conventional approaches to classifying brain tumors, key steps include skull stripping, defining regions of interest, segmenting features, and selecting features. However, with deep learning, the need for these preprocessing steps is mitigated as the model can autonomously extract relevant features, enabling the utilization of whole-brain MRI for decision-making without prior preprocessing [36].

CNN-Based Methods

Recent studies show that deep learning frameworks like convolutional neural networks, Mask-Recurrent Neural Networks, and UNET outperform conventional methods in segmentation [37]. In the proposed framework of Harvei et al. [38], the tumor is segmented using a fully automatic deep learning-based technique called InputCascadeCNN. A unique CNN architecture differs from other traditional CNNs due to its two-way processing of images. InputCascadeCNN uses a final layer that is a convolutional implementation of a fully connected layer. The architecture of InputCascadeCNN consists of two streams: one with 7×7 accessible fields for extracting local features and another with 13×13 accessible fields for extracting global features. The research article in [39] concentrated on a comparative analysis of various image segmentation algorithms employed in brain tumor diagnosis. The algorithms under consideration included Otsu's, watershed, level set, K-means, HAAR Discrete Wavelet Transform (DWT), and CNN. The simulations for these techniques were conducted in MATLAB using images sourced from the Brain Tumor Image Segmentation Benchmark (BRATS) dataset-2018. The measured accuracy for Otsu's, watershed, level set, K-means, DWT, and CNN methods was reported as 71.42%, 78.26%, 80.45%, 84.34%, 86.95%, and 91.39%, respectively. The CNN architecture consists of three layers with two sequential convolutional layers, pooling layers, and a third fully connected layer for classification. This architecture achieved the highest accuracy in less runtime compared to other methods used in the research.

Moreover, [1] proposed a segmentation model using Unet structure and 39 layers CNN. While [40] proposed MDFU-Net Multiscale dilated features upsampling network for segmentation of heterogeneous datasets. Sangui et al. [41] used U-Net architecture for 3D MRI segmentation. In [42], the authors developed a model for brain tumor classification using the RCNN technique. RCNN is an object detection mechanism based on CNN, which efficiently classifies image regions and generates the detected bounding boxes containing the region of interest (ROI). The proposed model used two-channel CNN to classify between glioma and healthy brain and to detect the tumor regions of the glioma. Therefore, Masood et al. suggested in [43] a proprietary Mask Region-based CNN with a dense net-41 backbone architecture that was learned using transfer learning. Similarly, [14] proposed an end-to-end, lightweight deep learning model for brain tumor classification into two classes (tumor or normal) for binary classification and into three classes (meningioma, glioma, and pituitary tumors) for multi-class classification. The model architecture consists of CNN with multiple convolutional and pooling layers, followed by fully connected layers and a softmax activation function for classification. The model was trained and evaluated on two publicly avail-

able brain tumor datasets using various performance metrics, including accuracy, sensitivity, specificity, and F1-score. The study also compared the proposed model's performance with other state-of-the-art models and demonstrated its superior performance in terms of accuracy and efficiency.

The CNNs explored in earlier articles faced notable limitations, struggling with insufficient power to effectively manage the complexities of deep multichannel, multilayer structures featuring a substantial number of parameters. Compounded by the constraint of relatively small datasets and the absence of key training techniques like parameter initialization heuristics and effective regularization, these models encountered challenges in optimization. Moreover, computational demands placed a significant burden on available resources. Recognizing these constraints, a promising avenue for overcoming these hurdles is the adoption of transfer learning. By capitalizing on pre-trained models, transfer learning allows the transfer of knowledge gained from one task to another, leveraging deep features learned from more robust datasets and powerful models. This approach proves particularly advantageous in scenarios where data is limited or computational resources are constrained, providing a strategic solution to enhance the overall effectiveness of convolutional neural networks (Fig. 12).

Deep Autoencoder-Based Methods

A deep autoencoder is an expansion of the ordinary autoencoder with the exception that DAE has more layers than a simple autoencoder and thus is able to learn more complex features and produces sensitive power. Gonbadi et al. employed a deep autoencoder in their work [44] in which the number of hidden layers for encoding and decoding parts of the DAE are decided based on input size. This approach achieved an accuracy of 97%. Raja [27] proposed a hybrid scheme of the DAE-based JOA (Jaya optimization algorithm) with a softmax regression technique.

The authors in [45] used a deep convolutional autoencoder for the segmentation of brain tumors. The CAE contains three blocks, the first is the encoder part, which includes convolution with a filter-bank, max-pooling, and sub-sampling to generate a set of feature-maps, the second part represents the bottleneck and the third is for the decoder part of the network. The developed network achieved an accuracy of 97%. Figure 14b represents the architecture of deep autoencoder-based classifier in [45].

However, [46] proposed a twin-attention-based dense convolutional auto-encoder for brain tumor classification. The twin-based attention mechanism includes feature-based and slice-based attention for brain tumor detection. This attention mechanism enhances accuracy and efficiency by focusing on the essential input elements. The autoencoder consists of four network layers: encoder, decoder, bottleneck layer, and

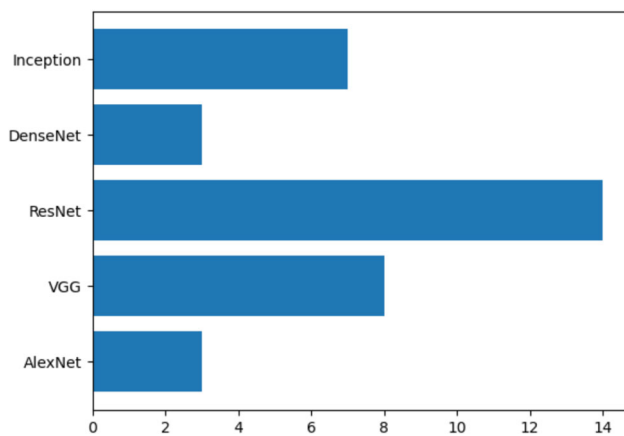


Fig. 12 Utilization of pre-trained models in brain tumor classification articles: distribution across various architectures (AlexNet, VGG, ResNet, DenseNet, Inception)

reconstruction loss unit. The proposed approach achieves an accuracy of 97.28%.

Transfer Learning-Based Methods

The authors in [11] merged 23 CNN layers with a VGG16 model in order to overcome the problem of overfitting faced in training a small dataset of 152 MRI images (as illustrated in Fig. 13a). To address the problem of imbalance datasets [30] designed a deep ensemble model composed of shallow CNN for features extraction and a fine-tuned deep learning of VGG16 model connected to three vectors of fully connected layers. Thus, VGG has limitations such as time-consuming and takes a lot of disk space with 138 million parameters which leads to the exploding gradient problem. Consequently, ResNet was introduced by many researchers to prevent this gradient problem. Authors in [47] proposed a multimodal-Link-Net composed of an encoder-decoder network and ResNet152 as a backbone discriminator for segmentation. Moreover, [43] developed Custom Mask Region-based CNN (Mask RCNN) with a Densenet-41 backbone architecture. In another paper [29], authors used an Inception v2 net hybridized with 16 layers with Fast Fuzzy C-means for the detection of tumor core and edema.

Therefore, [23] achieved that ResNet-50 has demonstrated exceptional performance in all evaluation measures compared to other models. Hence, the work demonstrated its efficacy in the detection of brain tumors with the allocation of resources; less energy, runtime, and memory use compared to other models under consideration. The same result was achieved in [50] in which the authors employed several transfer learning algorithms, including Xception, ResNet50V2, InceptionResNetV2, and DenseNet201. An efficient skip connections-based residual network for brain tumor classification has been developed in [51], addressing the limitations

of existing models by incorporating residual blocks with skip connections to enhance gradient flow during training. The architecture featured multiple stages with increasing numbers of residual blocks, promoting in-depth feature learning and pattern recognition, while also integrating efficient down-sampling techniques and batch normalization layers to enhance robustness and reliability. The model demonstrated exceptional performance metrics. However, potential limitations include the need for further clinical validation, considerations regarding interpretability, and the practical implementation of the model in real-world healthcare settings. Authors in [52] involved the development and fine-tuning of a deep learning model, specifically YOLOv7, for the purpose of detecting brain tumors in MRI scans. YOLOv7 is an extension of the YOLO (You Only Look Once) architecture and is known for its effectiveness in detecting objects in images. The authors fine-tuned the YOLOv7 model through transfer learning to improve its performance in identifying different types of brain tumors, including gliomas, meningiomas, and pituitary tumors, within MRI images. The authors also applied data augmentation techniques to the dataset, such as rotation and flipping, to enlarge the training set. Additionally, the authors used Albumentations and Keras normalizing functions for data preprocessing and normalization of MR images. Figure 12 shows the distribution and utilization frequency of pre-trained models across the articles in the study. It provides insights into the prevalence and adoption of these models in brain tumor detection and classification.

Transformers-Based Methods

The transformer architecture features attention mechanisms as a crucial component. Over recent years, attention mechanisms have gained rapid prominence in computer vision. Notably, Jun et al. identified limitations in CNNs for pinpointing the focal point of a lesion [53]. In their work, they proposed a novel brain tumor classification method based on an attention mechanism and a multipath network to address this challenge. The dataset comprises T1-weighted post-contrast brain images from Figshare, encompassing three types of tumor images: gliomas, meningiomas, and pituitary adenomas. The dataset includes 3064 image slices of size 512x512, expanded to 9192 images through data augmentation methods. The attention mechanism selectively extracts critical information related to the target region while filtering out irrelevant details. The multipath network is equivalent to grouping convolution. It assigns the data to multiple channels, before converting each channel and concatenating the results of all branches. Experimental evaluations of this model achieved an overall accuracy of 98.61%, which outperforms previous studies on the same dataset.

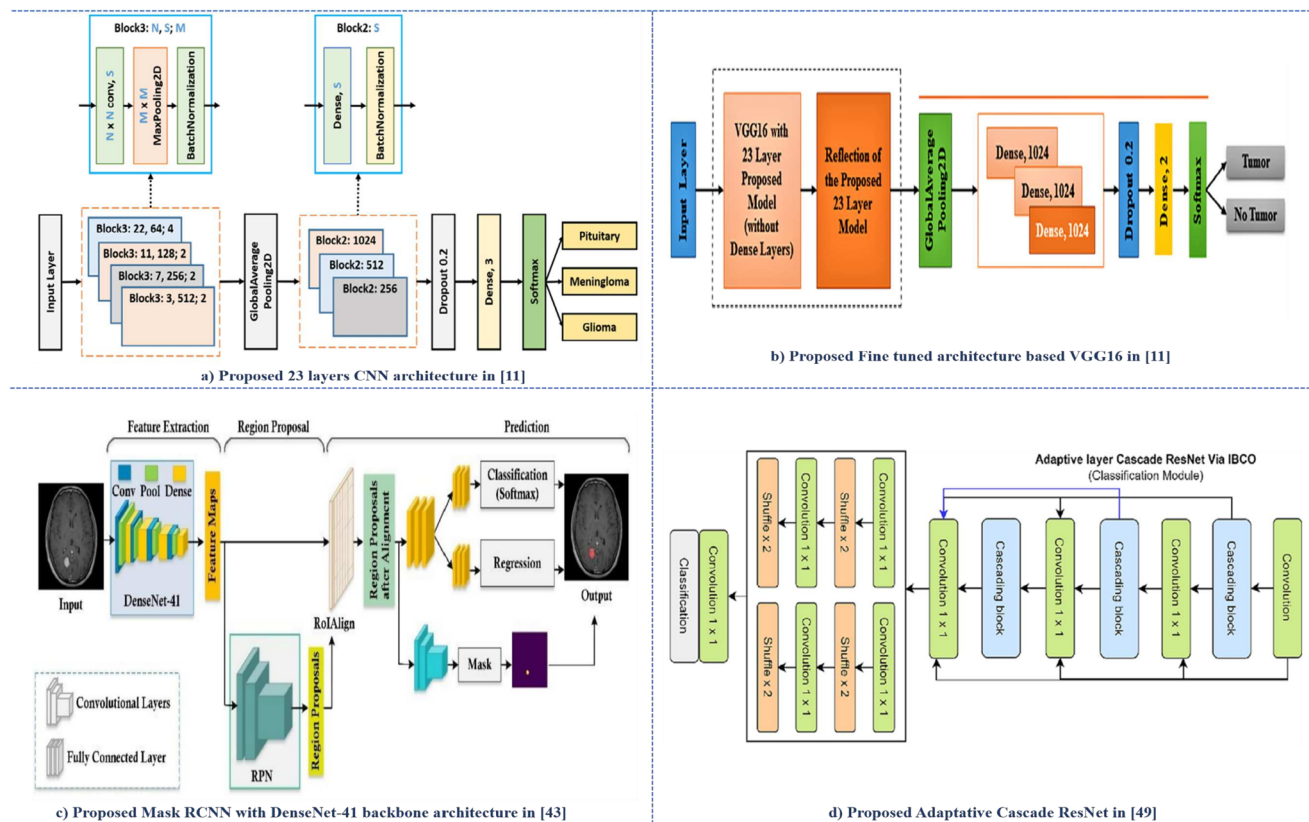


Fig. 13 Comparative visualization of four CNN architectures for brain tumor detection and classification: **a** illustrate the proposed 23-layer CNN architecture described in [11]. The network consists of four blocks of layers, leading to global average pooling. It then applies a block of densely connected layers with batch normalization, followed by a fully connected layer and a final layer for multiclass classification.

b Showcases the proposed fine-tuned architecture based on VGG16, as presented in [18] and optimized for binary classification tasks. **c** Reveals the architecture of the proposed Mask R-CNN with DenseNet 41 backbone, as described in [43]. **d** Presents the architecture of the proposed Adaptive Cascade ResNet in [49], which introduces adaptability through cascaded ResNet blocks

The study in [54] highlights meaningful features using a compact split-attention(CSA) u-shape network based on feature conservation and compact split-attention block, while [55] introduced a spinal convolution attention network using transformers. Alzahrani and Qahtani [56] used knowledge distillation from tripartite attention; temperature softening neighborhood attention, global attention, and cross-attention layers. Transformer and 3D CNN have been combined in [57] based on the encoder-decoder structure in an architecture called TransBTS. Therefore, [58] proposed a transformer model with convolutional layers along with self-attention and external attention called ConvAttenMixer.

In [59] authors introduced an attention-based residual multi-scale CNN designed for multi-classification of brain tumors. The model incorporates a lightweight residual multi-scale CNN to capture high-level feature representations across various receptive fields. Additionally, a lightweight global attention module is integrated to selectively learn discriminative features and explore extensive feature dependencies. Nevertheless, the model's robustness and parameter performance warrant comprehensive testing under diverse

conditions. Mishra and Verma [60] proposed a combination of graph attention autoencoder (GATE) and convolutional neural network (CNN) frameworks for brain tumor classification using MRI images. The GATE framework is used to convert the input MRI image into a graph format, where each pixel is represented as a node in the graph. The attention values of neighboring nodes are calculated for each node in the graph, which helps to capture the dependencies between neighboring pixels. The output of the GATE framework is then converted back into an image format and passed to CNN for further processing and classification. The CNN architecture used in the paper consists of two convolutional layers, two dense layers, and two dropout layers to prevent overfitting. The model is trained using the Adam optimizer with a learning rate of 0.01 and ReLU and Softmax activation functions. The performance of the model is evaluated using accuracy, loss, precision, F-score, and recall metrics on three different brain tumor datasets.

Moreover, a multi-objective optimization approach has been employed by [61], integrating both Single Objective Optimization (SOO) and Multi-Objective Optimization

(MOO) methods to enhance the architectural parameters of the Vision Transformer (ViT) model for efficient brain tumor classification from 2D MRI images, with a focus on accuracy and model size reduction. The BMO-optimized ViT model demonstrated a substantial improvement in accuracy, achieving a notable accuracy of 0.9809, showcasing the effectiveness of the multi-objective optimization approach in enhancing the model's performance for brain tumor classification. Figure 14c shows the architecture of the proposed model in [61].

Asiri et al. [62] used five fine-tuned vision transformers to classify brain tumor images. The ViT models are R50-ViT-I16, ViT-b16, ViT-I16, ViT-b32, and ViT-I32, each with different resolutions and architectures. The models are fine-tuned on the brain tumor dataset to improve their performance on this specific task. The study also compares the performance of the ViT models against various existing state-of-the-art techniques for classification, such as convolutional neural networks (CNNs) and transfer learning. Overall, the study demonstrates the effectiveness of ViT models in

accurately classifying brain tumor images and provides a benchmark for future research in this field.

Generative Adversarial Network GAN

Generative adversarial networks GAN is a deep learning model in which two neural networks compete to improve their prediction accuracy. A convolutional neural network serves as the generator, while a deconvolutional neural network serves as the discriminator. The generator's objective is to generate outputs that might easily be mis-interpreted as real data. The discriminator's purpose is to figure out which of the outputs it gets were generated intentionally. Ian Goodfellow et al. [63] first described the GAN architecture in 2014. GAN is an efficient generative modelling process that shows properties similar to the original dataset. The quality of samples is increased via GAN. Some data augmentation approaches alter the gathered images, including scaling, flipping, rotating, and translating. It enhances the quality of samples. The class imbalance issue can be solved via GAN.

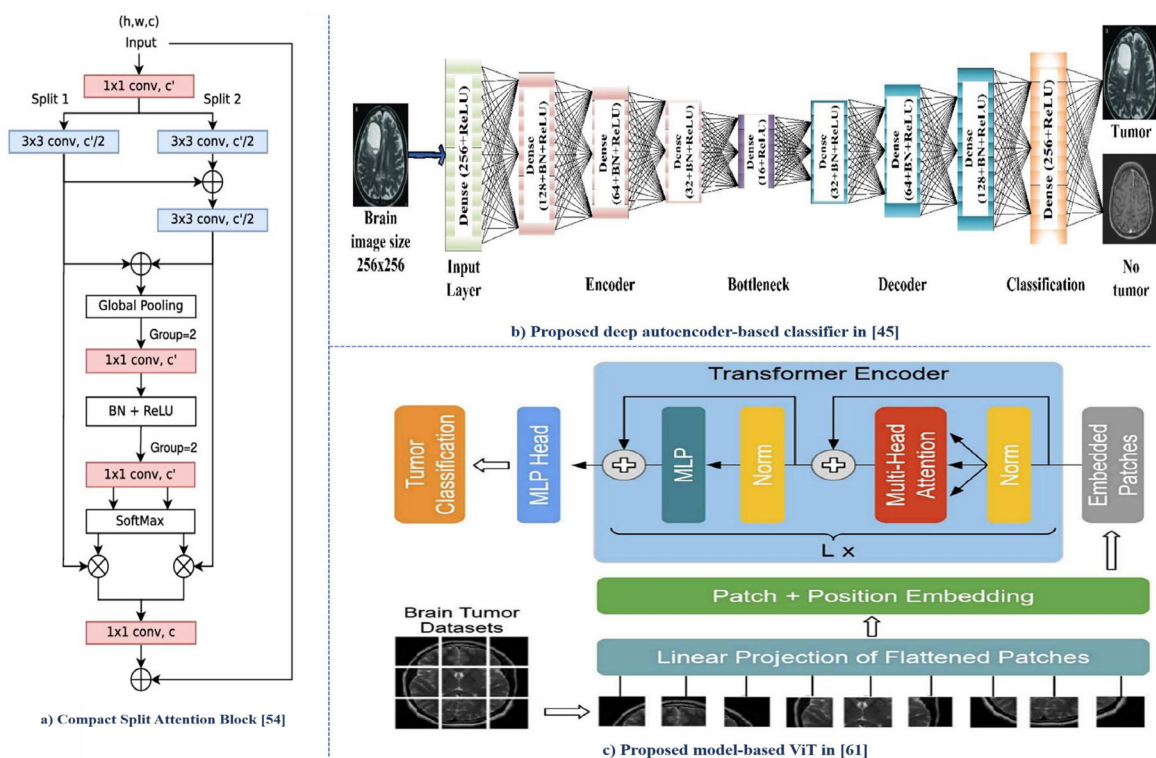


Fig. 14 Architectural overview of three innovative models: DAE, Transformers, and Attention Mechanism: **a** Showcases the Compact Split Attention Block, as introduced in [54]. This attention mechanism enhances feature representation by efficiently capturing spatial and channel-wise dependencies. **b** Illustrates the architecture of the proposed Deep Autoencoder-Based Classifier detailed in [45]. This model is tailored for binary classification tasks and encompasses distinct layers: an input layer, an encoder for feature extraction, a bottleneck layer

for latent representation, a decoder for reconstruction, and a binary classification layer. **c** Illustrates the innovative model based on the ViT architecture as outlined in [61]. Patches from the input image undergo linear projection and flattening before incorporating position embeddings. The transformed data then enters a transformer encoder, comprising normalization layers, multihead attention, a multi-layer perceptron, and final output for tumor classification

Researchers developed models using GAN due to the lack of large datasets of brain tumor MRI images. For instance, [49] used GAN to augment the data regarding contrast-enhanced images. Salama et al. in [64] developed a generative model by modifying the version of the variational autoencoder. The generative model captured the distribution of the important features in a set of small class-unbalanced brain MRI images and synthesized any number of samples for each class. Therefore, [22] suggested a deep convolutional Generative Adversarial Network that generates synthetic images that deceive the discriminator into pretending they are real followed by other preprocessing techniques.

In addition, a novel framework denoted BrainGAN was developed by [87]. The paper discusses the use of GAN architectures, specifically Deep Convolutional GAN (DCGAN) and Vanilla GAN as illustrated in Fig. 15, for the generation and classification of brain MRI images. These generated images are then validated using deep transfer learning models such as CNN, MobileNetV2, and ResNet152V2. BrainGAN involves training the deep transfer models with images created by GAN architectures and evaluating their performance on a test set comprising real brain MRI scans. However, the ResNet152V2 model demonstrated superior performance in terms of accuracy, precision, recall, area under the curve (AUC), and loss when evaluating images generated by the DCGAN architecture.

Comparative Analysis

Table 5 represents a comparative analysis of the reviewed articles. We used the forms provided in Appendix 1 to extract relevant information from the studies included in this review and summarized it in Table 5. This information includes the datasets and their sizes, the modalities used, the approach and algorithms used, as well as information about the environment configuration and software used. Moreover, the table shows the results of each research article in terms of accuracy, and we also mention some limitations and highlights of each research.

Limitations and Gaps

Related to MRI Images

MRI images can acquire artifacts such as artifacts stemming from patient movement, aliasing due to a small field of view, magnetic susceptibility issues, and interference from metallic or foreign objects such as tattoos, hairbands, clothing ornaments, endovascular embolization, prostheses,

orthodontic braces, neurosurgical clips, and other medical implants. Knowledge of the artifacts in MRI is extremely important because it may cause false diagnoses or reduce the quality of examinations leading to false positive classification or detection.

On the other hand, many researchers commonly work with datasets that exhibit imbalanced data, particularly when the groups to be classified have unequal voxel distributions. Pre-trained models often face challenges and fail to achieve satisfactory results when trained on such imbalanced datasets [11].

Therefore, existing methods designed for accurate brain tumor segmentation, which are based on homogeneous datasets, experience significant performance degradation in real clinical applications and lack the analysis of heterogeneous data [40]. Brain tumors typically consist of three heterogeneous parts that may or may not be connected. These parts vary in physical and chemical properties, including edema, necrotic tissue, and the enhanced tumor core. The heterogeneous appearance of these parts sometimes makes it difficult to predict whether the pixel under consideration belongs to the foreground or background [75].

For instance, distinguishing between high-grade and low-grade (grade 2 and grade 3) tumors may present difficulties due to their limited inter-class variability, leading to reduced accuracy when utilizing extracted features for classification. This highlights the importance of addressing interclass variability in developing effective models for tasks such as brain tumor segmentation [49].

Related to Datasets

Significant demand exists for freely available datasets in medical research, particularly in fields such as image analysis and segmentation. Many researchers rely on open-access repositories, with Figshare and BraTS being among the most frequently utilized platforms. However, these datasets often come with limitations, including relatively small sample sizes, imbalanced data distributions, and the presence of extreme noise and artifacts. Moreover, a noteworthy challenge arises from the fact that these datasets are not consistently oriented in a single plane. They encompass mixed planes, including sagittal, axial, and coronal views, which deviate from the standardized anatomical structure and spatial orientation. For instance, Figshare lacks adherence to a standard format, making it essential for researchers to be mindful of these complexities when using such datasets for training and evaluation in medical image analysis tasks.

In addition, another noteworthy aspect of datasets used in medical research is the variation in imaging modalities. These

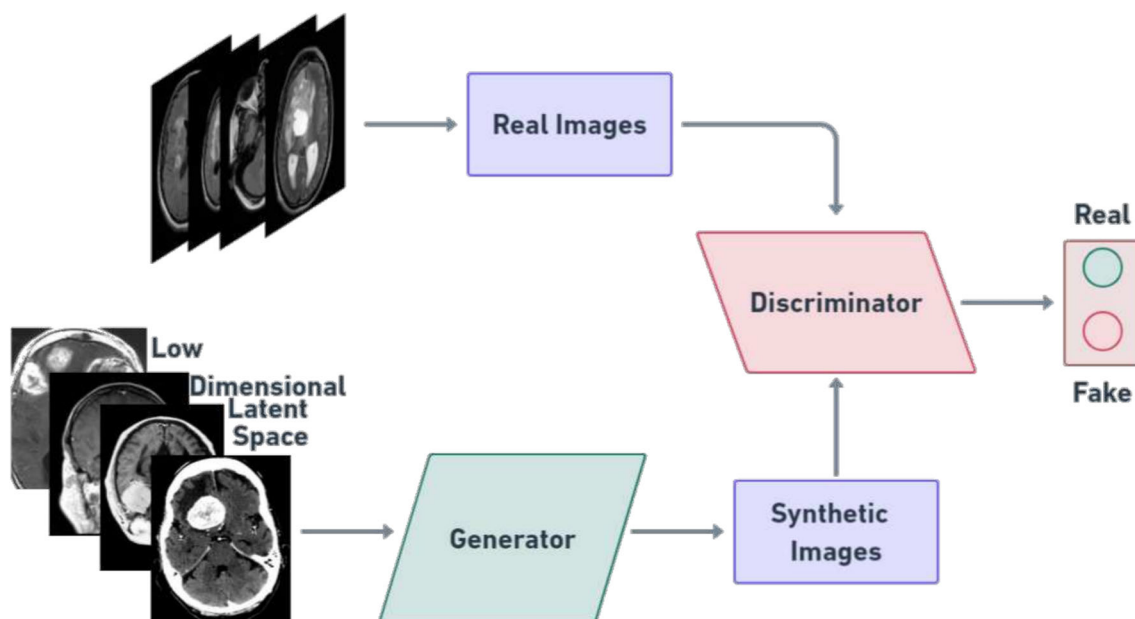


Fig. 15 Vanilla GAN architecture [87]. The generator network takes an input image and uses random noise to generate synthetic visuals. At the same time, the discriminator network receives inputs from two sources: the original images from the real dataset and the images generated by the generator

datasets often incorporate multiple modalities, including T2, T1, and T1 contrast-enhanced, all within a single dataset. While this diversity provides comprehensive information, it introduces practical concerns. The signal characteristics of tumors can vary significantly across modalities, potentially leading to challenges in model generalization. For instance, the signal intensity of a tumor may differ between T2, T1, and T1 CE modalities, making it essential for researchers to carefully consider the implications of this modality heterogeneity during training and evaluation. Addressing these multifaceted challenges is crucial for developing robust and reliable models in the field of medical image analysis.

On the other hand, the employment of data augmentation techniques proves highly beneficial for working with an extensive dataset, allowing models to operate efficiently and achieve high-performance metrics. Nevertheless, a notable drawback of some data augmentation approaches is their potential to contribute to false diagnoses. Certain techniques, such as scaling, flipping, rotating, and translating, alter the acquired images, enhancing the overall quality of the samples [49]. However, when these techniques involve extensive flipping and rotating at various angles, there is a risk of training the model in a manner that does not preserve the normal anatomical positions.

While researchers have introduced various algorithms to increase data volume, the primary objective is often to mitigate overfitting in the training dataset. Unfortunately, these approaches frequently overlook the consideration of spatial

and textural relationships. This underscores the necessity for a standardized augmentation technique to facilitate comparative analyses. Generative Adversarial Networks have been employed to address this concern, aiming to enhance spatial and textural fidelity. However, GANs may still encounter challenges in terms of performance and occasionally generate synthetic, unrealistic images. Striking a balance between augmenting data for improved model performance and maintaining anatomical accuracy remains a crucial consideration in the development of robust medical image analysis models.

Related to CNN Models

The inherent complexity of pre-trained models can present challenges in the interpretation and understanding of the specific features that influence classification [71]. This complexity underscores the necessity for further clinical validation, considerations regarding interpretability, and practical implementation of the model in real-world healthcare settings. Although, these models face challenges such as restricted depth, issues with vanishing gradients, and difficulties in capturing intricate features [51].

Furthermore, the lack of interpretability in CNNs, especially deep architectures adds to the difficulty in comprehending their decision-making processes and becomes a notable concern in medical diagnosis. In addition to interpretability challenges, another drawback is the problem associated

Table 5 Comparative analysis of literature reviews

Datasets	Ref	Year	Approach	Algorithm	Metrics	Sample size	MRI sequences	Environment configuration	Limitations	Highlights
Figshare Kaggle	[11]	2023	Deep CNN classification and segmentation using U-Net structure	Segmentation using U-Net structure and 39 layers CNN	Accuracy = 98.8%	3064 images 17556 images	T1-CE, T1, T2, and FLAIR	Google colab and using personal computing devices	Not applied in real-life MRI data for segmentation and training	Evaluated across multiple and large datasets with five pre-trained models
Figshare Kaggle	[42]	2021	Classification and detection using the RCNN technique	Two channels CNN used as a classifier and feature extractor of RCNN	Accuracy = 98.21%	3064 images 1426 images	T1-CE	MATLAB 2020 8 GB RAM and 2.6 GHz clock frequency	Limited to object detection	Less execution time
Harvard Medical Figshare	[11]	2022	Binary classification and multi-class using two deep CNN	23 layers CNN and Fine-tuned VGG16 architecture	Accuracy = 100%, 97.8%	152 images 3064 images	T1-CE	Google colab	Limited datasets and the work is not validated on actual clinical study	Promising classification outcomes
BRATS 2015	[27]	2020	Classification using a hybrid deep autoencoder	BFC for segmentation. DAE + JOA + Softmax regression for classification	Accuracy = 98.5%	Not mentioned	T1-CE, T1, T2 and FLAIR	Matlab	Complex architecture	More focusing in pre-processing and features extraction techniques
BRATS 2020	[65]	2022	Binary classification using CNN	Convolutional neural network followed by SVM classifiers	Accuracy of 99.74%	2892 images	T1-CE, T1, T2 and FLAIR	Not mentioned	Limited datasets	Less parameters
Figshare Kaggle	[43]	2021	Recognition and Classification using deep learning	Custom Mask Region-based CNN (Mask RCNN) with a denseNet-41 backbone architecture	Accuracy segmentation 96.3%, Classification 98.34%	3064 images 253 images	T1-CE	Not mentioned	Limited datasets	Custom Mask-RCNN can compute deep features with effective representations
Figshare	[53]	2022	Classification based on attention mechanism and a multipath network	Dual-attention blocks generated by adding Se module and gated channel transformation to the multipath block Xception ResNet50V2 InceptionResNetV2 DenseNet201	Accuracy = 98.61% 99.4% 99.68% 99.36% 98.72%	3064 images	T1-CE	GPU Titan V 12 G Cuda 10.2	Misclassification for meningioma samples and overfitting due to limited datasets	Solves the challenge of a large number of parameters for limited computing resources
Figshare	[50]	2023	Classification using pre-trained model					Intel Xeon CPU with 2 Cores, 13 GB RAM, and 16 GB GPU MATLAB	Limited datasets	High classification accuracies
Figshare	[17]	2022	Fully automatic detection and segmentation (Meningioma)	Geometric segmentation algorithm: level set techniques Localisation: pre-trained CNN	Dice similarity coefficient = 0.95	485 meningioma images in axial, coronal, and sagittal	T1-CE		Tumors might require a careful consideration of the shape and structure of the tumors to be detected and segmented	Require a relatively light training by exploiting the potential of the transfer learning approach

Table 5 continued

Datasets	Ref	Year	Approach	Algorithm	Metrics	Sample size	MRI sequences	Environment configuration	Limitations	Highlights
BraTS 2017 BraTS 2020	[29]	2023	Hybrid DNN for multimodal brain tumor segmentation	Inception v2 net hybridized with 16 layers Fast Fuzzy C-means for detection of tumor core and edema	Accuracy = 99.45%, DSC = 89.74%	1140 images	T1-CE, T1, T2 and FLAIR	MATLAB 2018	Limited datasets	combined advantages of inception v2 net and Fast fuzzy C-means method for tumor segmentation
Figshare BRATS 2018	[49]	2023	Automated classification and grading systems	Generative model of GAN for Data Augmentation, Binary and multiclass classification using ALCResNet optimized by IBCO algorithm and DeepLabV3 for segmentation of abnormal images	Accuracy = 99.51%	3064 images 233 images	T1-CE, T1, T2 and FLAIR	Not mentioned	Complex architecture with high computational resources	Improve accuracy, computational time, and convergence performance for brain tumor diagnosis and grading
Figshare	[30]	2023	Detection using ensemble deep CNN	Shallow CNN for feature extraction and VGG16 model	Accuracy = 97.77%	3064 images	T1-CE	NVIDIA K80 GPU with 25 GB RAM	Limited datasets	High-level information about the tumor localization
BraTS 2020	[40]	2023	Segmentation from heterogeneous data using MDFU-Net	MDFU-Net Multiscale dilated features upsampling network	Accuracy homogeneous: 82.96%, Heterogeneous: 62.66% IoU = 93.6% IoU = 89.3% IoU = 93%	Not mentioned	T1-CE	NVIDIA K80 GPU with 25 GB RAM	The diagnostic performance for the heterogeneous brain dataset is limited	No previous research that considers the complete heterogeneous brain data analysis
Kaggle	[23]	2023	Classification using hybrid CNN architecture	ResNet-50 VGG16 DenseNet Inception V3	IoU = 71.8%	3929 images	T1-CE	Intel's developer cloud and the oneAPI toolset	Limited datasets	Model optimized with oneAPI consequently more comprehensive and discriminative features
BraTS 2020	[47]	2023	Multi-Modal semantic segmentation model using Encoder Based Link-Net Architecture	MM-Link-Net: Encoder-decoder network and ResNet152 as backbone discriminator for segmentation	Accuracy = 99.2%	494 images	T1-CE, T1, T2 and FLAIR	GeForce GTX TITAN Black of 6GB RAM	Computationally expensive and time-consuming because of the depth of layers in architecture and separation of encoders	The proposed model performs well in segmentation
BraTS 2021	[54]	2022	Segmentation using split-attention UNet	Encoder-decoder: Compact Split Attention UNet with primary feature conservation strategy	Accuracy = 0.935% ± 0.015	1251 images	T1-CE, T1, T2, and FLAIR	PyTorch 1.10.0 framework on a single NVIDIA V100 Tensor Core GPU	Limited datasets	Alleviate gradient vanishing problem with increasing layers, Improves computational efficiency, Extends the receptive field of the network

Table 5 continued

Datasets	Ref	Year	Approach	Algorithm	Metrics	Sample size	MRI sequences	Environment configuration	Limitations	Highlights
BraTS 2020	[41]	2023	3D MRI Segmentation using U-Net Architecture	U-Net architecture: the encoder part's layers are skip connected and are concatenated with the decoder part's layers	Accuracy = 99.4%	371 training files	T1-CE, T1, T2, and FLAIR	2-core of Intel Xeon as CPU, Tesla P100 16GB VRAM as GPU, with 13GB RAM	Limited datasets and highly time consuming	Enhances the level of image segmentation and its spatial localization
Kaggle	[64]	2020	Detection based on convolutional variational generative models	Generative model capturing features in small class-unbalanced brain MRI images and a classifier based on CNN	Accuracy = 96.88%	253 images	T1-CE, T1, T2, and FLAIR	Google colab	Highly time-consuming	The system automatically converts a small unbalanced dataset to a larger balanced one
Figshare	[48]	2022	Recognition using an integrated bat algorithm with a CNN approach	CNN and Bat algorithm for feature extraction and the 2-D Gabor filter for feature selection	Accuracy = 99.5%	3064 images	T1-CE	Google Colab	Limited datasets	Less complex time, and higher processing speed
Figshare BraTS 2017 and others	[45]	2022	Classification using deep convolutional autoencoder DCAE	Spectral Data Augmentation based Deep Autoencoder	Accuracy = 97%	3000 images	Not mentioned	Intel i5 8th Gen, 2.4 GHz processor, 16GB RAM	Limited datasets	Model perform high with spectral data augmentation
BraTS 2019	[55]	2023	Binary classification (malignant and benign) using Gradient awareness minimization-based spinal convolution attention network (GAM-SpCaNet)	Spinal Convolution Attention Network. Positional Attention (PA) convolution block, Relative self-attention transformer block, and Intermittent fully connected (IFC) layer	Accuracy = 99.28%	3040 images	T1-CE, T1, T2, and FLAIR	NVIDIA QUADRO RTX 8000	Hardware resource requirements	More lightweight and efficient in recognition of brain tumors
Kaggle	[56]	2024	Multiclass brain tumor detection	Knowledge distillation from tripartite attention: temperature-softening neighborhood attention, global attention, cross-attention layers	Accuracy (teacher) = 80.07% Accuracy (Student) = 76.22%	4455 images (BTDS-15C), 4415 images (BTDS-17C), 4478 images (BTDS-44C)	T1-CE, T1, T2 and FLAIR	NVIDIA A100 GPU courtesy of Google Colab Pro+	Hardware resource requirements	Highly augmented datasets and faster training times
BraTS 2019 BraTS 2020	[57]	2021	Multimodal brain tumor segmentation	TransBTS: Transformer in 3D CNN based on the encoder-decoder structure	Dice score = 90.09%	460 images (BraTS 2019) 494 images (BraTS 2020)	T1-CE, T1, T2 and FLAIR	Pytorch and trained with 8 NVIDIA Titan RTX GPUs (each has 24GB memory)	Limited datasets	3D CNN for modeling local context information and learning global semantic correlations. Trained from scratch

Table 5 continued

Datasets	Ref	Year	Approach	Algorithm	Metrics	Sample size	MRI sequences	Environment configuration	Limitations	Highlights
Figshare Kaggle	[58]	2023	Detection and type classification	ConvAttenMixer: transformer model with convolutional layers along with two attention mechanisms: self-attention and external attention	Accuracy = 97.94%	7023 images	T1-CE, T1, T2, and FLAIR	Python 3 Google Compute Engine backend (GPU) Type A100	Acquire more computational resources	Increased performance
				GoogleNet	Acc=96.08%					
				AlexNet	Acc= 95.16%					
				ShufNet	Acc=96.84%					
				SqueezeNet	Acc=97.17%					
Figshare	[66]	2023	Classification via hybrid deep learning techniques	NASNet-Mobile	Acc=97.50%	3064 images	T1-CE	Matlab 2023 GTX1050Ti (4GB)	Limited datasets	Hybrid DL technique using minimal pre-processing Several metrics are used to assess the system's performance
BrATS 2015	[67]	2021	Multiscale-based multimodal image classification using deep learning method	Multimodal Convolutional Neural Network with Long Short-Term Memory (MSMCNN-LSTM) and U-Net for Semantic Segmentation. Fivefold cross-validation scheme	Accuracy= 96.36%	274 images	T1-CE, T1, T2, and FLAIR	Not mentioned	Limited datasets	Segregate the classes of tumors by analyzing every pixel in the image
Figshare Kaggle	[68]	2023	Classification using CNN	Modified VGG-16 architecture	Accuracy= 99.5%	6328 images	T1-CE, T1, T2, and FLAIR	Google Collab Frameworks TensorFlow and Keras	Acquire more computational resources	Considerably higher accuracy
Figshare Kaggle	[69]	2023	Classification and detection via hybrid neural network architecture	AlexNet combined to GRU (Gated Recurrent Unit)	Accuracy= 97%	7023 images	T1-CE, T1, T2, and FLAIR	Tensor flow-based Python tool	Misclassification for some samples	Resolved gradient vanishing in deep networks using the GRU component
kaggle	[70]	2023	Binary Classification using IRNetv model	InceptionResNet backbone architecture	DS1 99.78 % DS2 99.14%	DS1: 4600 images DS2: 253 images	T1-CE, T1, T2, and FLAIR	Google Colab Python with Tensorflow and Keras libraries	No clear information about time consuming	Residual connections helped the model tackle the vanishing gradient problem

Table 5 continued

Datasets	Ref	Year	Approach	Algorithm	Metrics	Sample size	MRI sequences	Environment configuration	Limitations	Highlights
Figshare Kaggle	[71]	2023	Multiclass classification using Particle swarm optimized kernel Extreme Learning Machine (PSO-KELM)	Pre-processing: Guided filter and contrast enhancement Segmentation: Thresholding and other morphological operations Feature extraction: DenseNet201 and InceptionV3 Classification: Extreme Learning machine using Particle Swarm optimization	Classification accuracy DS1 97.92% DS2 98.21%	DS1:3264 images DS2:3046images	T1-CE	Python 3.10.9 version	Inherent complexity of Inception V3 and DenseNet201 might pose challenges in interpreting and understanding the specific features influencing the classification	Improved classification accuracy
Figshare Kaggle	[14]	2023	Binary and multiclass classification using a lightweight end-to-end deep learning model	End-to-end learning using CNN	99.48% for binary classification and 96.86% for multi-class classification	DS1 3064 images DS2 253 images	T1-CE	Google Colab	Limited datasets and the work is not validated on actual clinical study	Eliminates the need for traditional pre-processing, feature extraction, and separate classification
Figshare Kaggle	[52]	2023	Detection of brain tumor using You Only Look Once version 7 (YOLOv7)	Convolutional Block Attention Module (CBAM) attention mechanism included into YOLOv7	Acc=99.5%	10288 images	T1-CE	Nvidia GeForce 1080Ti GPUs, 32 GB RAM	Ambiguous dataset with duplicate instances	Based real-time object detection model
Figshare	[72]	2023	Brain tumor detection using deep learning and metaheuristic algorithm	Mobilenetv2 optimized by the Contracted Fox Optimization Algorithm (MN-V2/CFO)	accuracy=97.32 %	3064 images	T1-CE	NVIDIA GeForce GTX 1080 Ti graphics card, with 11 GB of GDDR5X memory and CUDA cores	Limited dataset and not validated on actual clinical study	Improved accuracy
Kaggle	[73]	2022	Multiclass classification using Recurrent CNN	Preprocessing: adaptive filtering algorithm, Segmentation: Clustering algorithm Feature extraction: GLCM Classification: RCNN	Accuracy=95.17%	3264 images	T1-CE	Keras libraries in python	Limited datasets, not validated on actual clinical study and lower accuracy	RCNN compared to U-Net and BP achieved the highest accuracy
TCIA (REM-BRANDT)	[36]	2023	Binary classification (LGG and HGG) using ensemble deep learning	MajVot (AlexNet, VGG16, ResNet18, GoogleNet, ResNet50)	FLAIR 98.88% T2 W 97.98% T1 W 94.75%	1430 images 897 images 1240 images	FLAIR, T1 W T2 W	Matalab2021b i5 processor with 4 GB NVIDIA	Absence of feature integration from multiple deep learning models	The study treats and provides accuracy for each sequence independently

Table 5 continued

Datasets	Ref	Year	Approach	Algorithm	Metrics	Sample size	MRI sequences	Environment configuration	Limitations	Highlights
Figshare	[51]	2023	Multiclass classification	Efficient Skip Connections-Based Residual Network (ESR-Net)	Accuracy= 99.62%	3064 images	T1-CE	MATLAB 2023a NVIDIA RTX A4000 Laptop GPU with 8 GB	Limited datasets, not validated on actual clinical study	Addressing the vanishing gradient problem
BraTS 2020 Figshare	[59]	2023	Multiclass classification using attention guided	Lightweight attention-based residual multiscale CNN ARM-NET	Figshare DS Acc= 96.64% BraTS DS Acc=97.11%	3064 images 396 cases	T1-CE, T1, T2 and FLAIR	Keras framework with Tensorflow backend	ARM-NET and the LGAM demands a few additional parameters	Superior performance lightweight and End-to-End Learnable
Kaggle	[60]	2022	Binary classification	Graph attention autoencoder (GATE) and convolutional neural network (CNN)	98.27% DS1 99.83% DS2 and 98.78% DS3	DS1 253 images DS2 1800 images DS3 971 images	Not mentioned	Google colab using Keras and Tensorflow python libraries	Limited dataset and computational complexity	Comparative analysis
Figshare Kaggle	[74]	2023	Multiclass classification	Transformer module (TM) with the self-attention unit (SAU) and convolutional neural networks (CNN)	Acc= 99.06%	3264 images	T1-CE	MATLAB 2022a	Limited dataset	Provide accuracy of iVGG, iDensNet, and iResNet and comparison with state of art
Figshare Kaggle	[61]	2023	Multiclass classification using ViT architecture	Bayesian Multi-Objective (BMO) optimization technique applied to ViT architecture	Accuracy = 98%	7023 images	T1-CE, T1, T2 and FLAIR	RTX 3060 GPU and RTX A6000 GPU	Acquire more computational resources and time consuming	Integration of BMO optimization with the ViT architecture results in a significantly improved model
Kaggle	[62]	2023	Multiclass classification using Fine-Tuned Vision Transformers	R50-ViT-I16 ViT-I16 ViT-I32 ViT-b16 ViT-b32	90.31% 97.89% 97.08% 94.51% 98.24%	5712 images	T1-CE, T1, T2 and FLAIR	Kaggle platform with 15.9 GB of GPU memory	Generalization of diverse medical images and cases needs further exploration	Comparative study of cutting-edge models using fine-tuning approach
Figshare	[46]	2024	Multiclass classification using twin-attention-based dense convolutional auto-encoder	Ganet-based Kapurs Thresholding for segmentation PHOG-GLRLM for feature extraction TA-CAE for classification	Accuracy= 97.28%	3046 images	T1-CE	Python Intel(R) Core(TM) i3-7100 CPU	Limited datasets and the work is not validated on actual clinical study	Better performance compared to other techniques of autoencoder

with pooling operations in CNNs. While pooling operations reduce the number of parameters in the model, they may simultaneously eliminate features that could be crucial to the overall system's efficiency. Current systems often lack intelligence in feature selection, relying on pooling techniques that, although beneficial for parameter reduction, can result in the removal of features that are essential entities for the holistic performance of the system.

General Challenges

General limitations include substantial hardware requirements, with models often consisting of millions or even trillions of parameters, necessitating the use of powerful GPUs. As a result, the need for high memory capacity further contributes to the cost, creating financial constraints and limiting the scalability of projects.

Therefore, researchers predominantly focus on classification tasks rather than detection, potentially overlooking critical aspects of real-world applications. Additionally, the optimization process poses challenges, as finding effective techniques and fine-tuning models can be time-consuming, impacting the efficiency of the overall research and development pipeline. These constraints highlight the importance of addressing hardware scalability, diversifying research objectives, and optimizing techniques to enhance the practicality and applicability of deep learning models.

Conclusion and Future Direction

In future research directions, the integration of multimodal data including MRI, CT, and PET scans could enhance the performance of deep learning models for brain tumor detec-

tion and classification. Thus, investigating transfer learning approaches to adapt pre-trained models may improve precision with limited labeled data. Moreover, prioritizing the development of the interpretability of models is essential for gaining acceptance within the medical community. In fact, exploring real-time applications of deep learning algorithms could have an immediate impact on clinical decision-making. On the other hand, encouraging multi-center collaborations for clinical validation studies will establish the generalizability of models. However, tackling challenges related to heterogeneous datasets, incorporating domain-specific knowledge and 3D images, handling imbalanced datasets, and conducting longitudinal studies will further advance the field and enhance the clinical relevance of deep learning in brain tumor detection and classification.

This systematic review of 60 research articles from 2020 to 2024 highlights significant advancements in deep learning for brain tumor detection and classification. While these algorithms show improved accuracy and efficiency in medical imaging analysis, challenges persist in model interpretability, data heterogeneity, and computational resource management. Future research should focus on addressing these issues through interdisciplinary collaboration and developing more interpretable, resource-efficient models capable of handling diverse datasets. Additionally, efforts to integrate these technologies into clinical workflows are crucial. This review offers valuable perspectives for researchers and practitioners, emphasizing the potential of deep learning to substantially impact neuro-oncology patient care and outcomes.

Appendix 1. Data Extraction Form

#	Extraction element	Contents
General information		
1	ID	Unique ID for the study
2	Title	Full title of the article
3	Authors	The authors of the article
4	Year	The publication year
5	Journal	The journal name, ranking and impact factor
Study description		
6	Study approach	☐ Tumor Detection ☐ Tumor Classification
7	Datasets	The source of the dataset used Size, types and modalities of images
7	Deep learning algorithms	☐ CNN ☐ TL ☐ DAE ☐ Transformers ☐ GAN ☐ Others
8	Metrics performance	The accuracy or other metrics of the proposed models
9	Environment configuration	Hardware or software used for implementation
10	Details about the study	Limitations and highlights

Fig. 16 Data extraction form

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Declarations

Consent to Participate All the authors involved have agreed to participate in this submitted article.

Consent for Publication The selected pictures of sample cases have personal consent form of each patient.

Conflict of Interest The authors declare no competing interests.

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